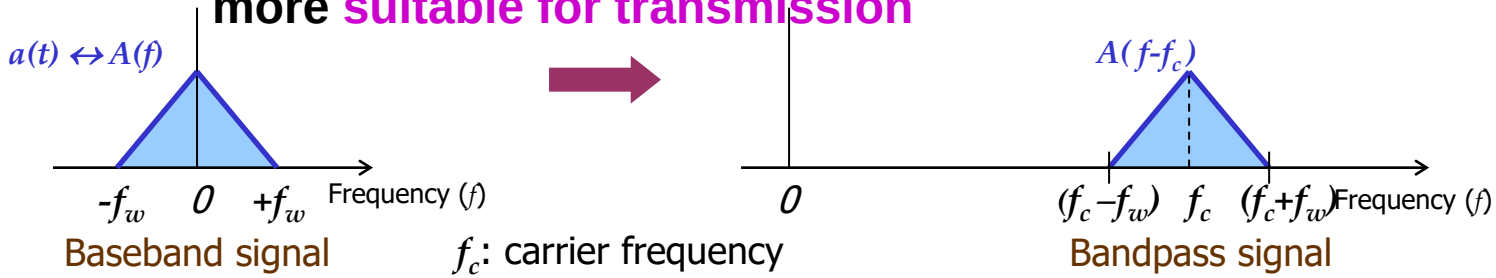


Update

- We have considered
  - Random process and signals
- We now consider
  - **Analog modulation** techniques
  - Amplitude modulation
    - DSB-LC, DSB-SC, QAM, SSB, VSB
  - Frequency division multiplexing (FDM)

Modulation Techniques

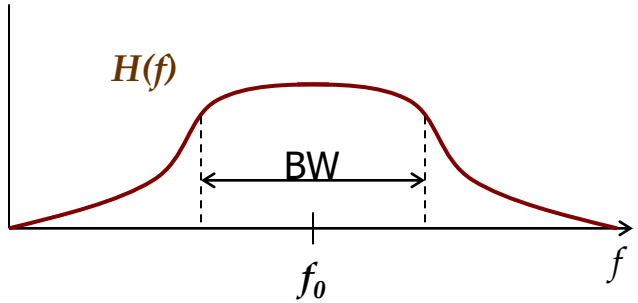
Modulation: -translates an information-bearing signal (message signal) to a new spectral location (frequency domain)
-a process that an original signal, $x(t)$, is used to alter a particular property of another signal (e.g. its amplitude, frequency, or phase), such that the resultant signal, $s(t)$, is more suitable for transmission



Baseband signal f_c: carrier frequency Bandpass signal

- Communication channels → bandpass transfer (frequency) response

  ⇒ translates the message signal to be within the channel transfer response



- **Facilitates antenna transmission/reception**

Selected frequency bands

Frequency Band	Carrier frequency	Bandwidth
Longwave Radio	100kHz	2kHz
Shortwave Radio	5MHz	100kHz
VHF	100MHz	2MHz
Microwave	5GHz	100MHz
Millimeterwave	100GHz	2GHz
Optical	$5x10^{14}$ Hz	10^{13}Hz

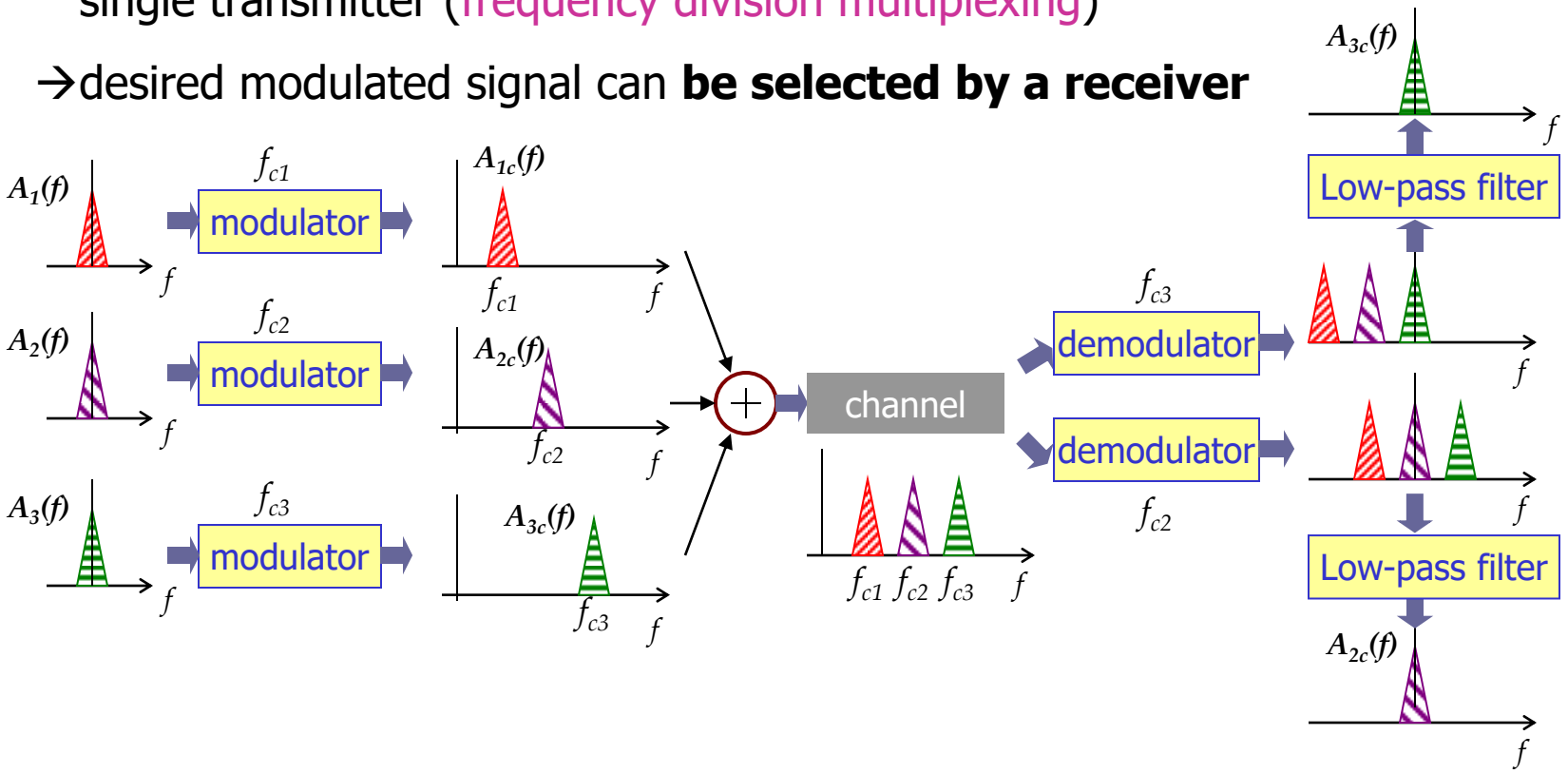
Modulation Techniques

If more than one message signal utilizes a channel

→**modulation** allows translation of different signals to different spectral locations

→**multiplexing** allows two or more message signals to be transmitted by a single transmitter (frequency division multiplexing)

→desired modulated signal can **be selected by a receiver**

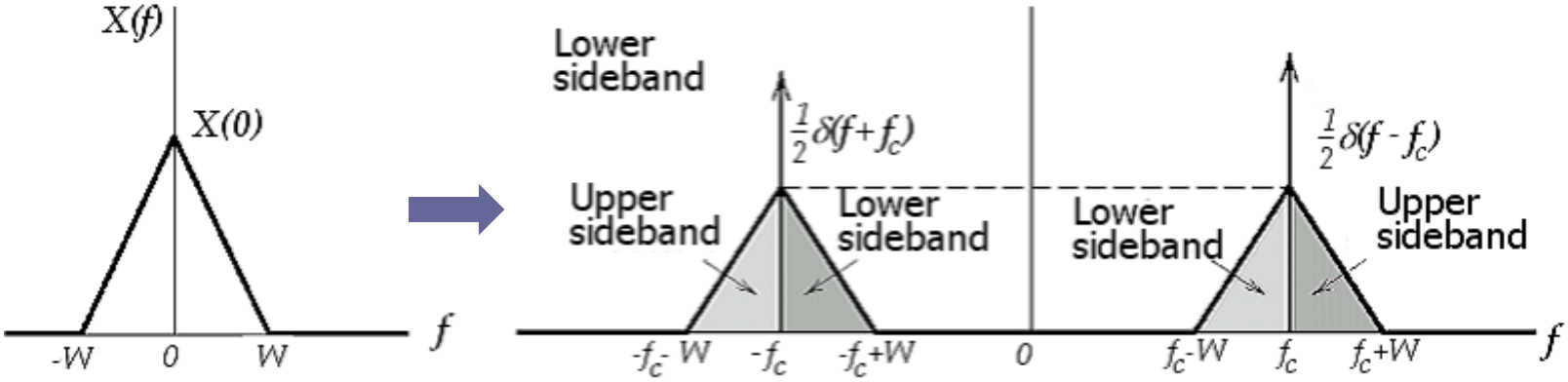


Frequency Translation

Recall: If $a(t) \leftrightarrow A(f)$, then $a(t)e^{j2\pi f_c t} \leftrightarrow A(f - f_c)$ (**Fourier transform pairs**)

Consider a **message signal** $x(t)$, which is bandlimited to the frequency range 0 to W and has its Fourier transform is $X(f)$, is multiplied by $\cos(2\pi f_c t)$.

$$\Im\{x(t)\cos(2\pi f_c t)\} = \frac{1}{2}[X(f + f_c) + X(f - f_c)]$$



Baseband signal

Frequency translated signal

Recovery of Baseband Signal

To **recover** the baseband signal, we can simply multiply the translated signal with $\cos(2\pi f_c t)$.

$$\underbrace{[x(t)\cos(2\pi f_c t)]}_{\text{Frequency translated signal}}\cos(2\pi f_c t) = x(t)\cos^2(2\pi f_c t) = \frac{x(t)}{2} + \frac{x(t)}{2}\cos(4\pi f_c t)$$

$$\updownarrow$$

$$\underbrace{\frac{1}{2}X(f) + \frac{1}{4}[X(f-2f_c) + X(f+2f_c)]}_{\text{Baseband signal}}$$

We **obtain the baseband signal** $x(t)$ and a signal whose spectral range extends from $(2f_c\text{-}W)$ to $(2f_c\text{+}W)$. As $f_c \gg W$, the extra signal is removed by a **low-pass filter**.

Analog (Continuous-Wave) Modulation

A parameter of a high-frequency **sinusoidal carrier** is varied proportionally to the **message signal** $x(t)$.

General modulated signal: $s(t) = A(t)\cos[\omega_c t + \phi(t)]$

ω_c : carrier frequency

$A(t)$: instantaneous amplitude

$\phi(t)$: instantaneous phase deviation

- When $A(t)$ is linearly related to the modulating (message) signal

  → **Amplitude modulation (AM)** $s(t) = A(t)\cos[\omega_c t + \phi_o]$ $where$ $A(t) \propto x(t)$

- When $\phi(t)$ is linearly related to the modulating signal

  → **Phase modulation (PM)** $s(t) = A_c \cos[\omega_c t + \phi(t)]$ $where$ $\phi(t) \propto x(t)$

- When *time* ***derivative of*** $\phi(t)$ is linearly related to the modulating signal

  → **Frequency modulation (FM)** $s(t) = A_c \cos[\omega_c t + \phi(t)]$ $where$ $\frac{d\phi(t)}{dt} \propto x(t)$

* FM & PM are commonly called **Angle Modulation**

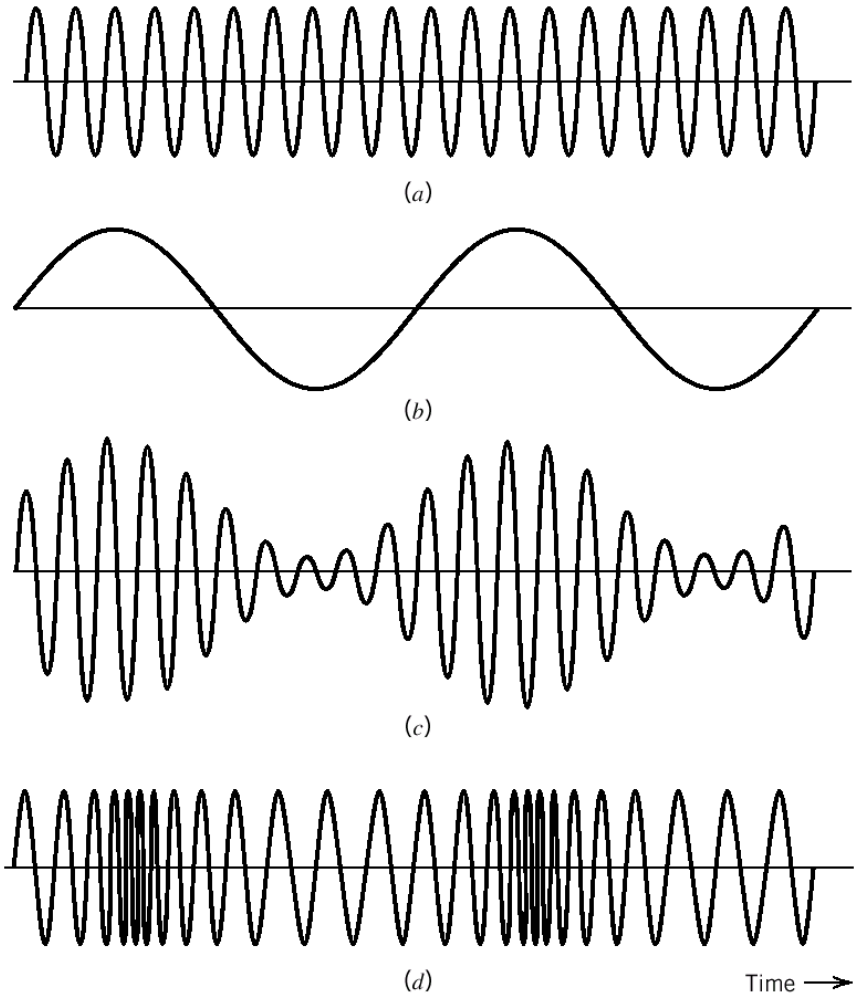
Analog (Continuous-Wave) Modulation

Unmodulated carrier frequency

Message signal

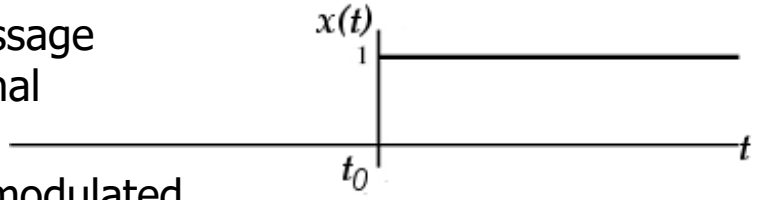
Amplitude-modulated signal

Angle-modulated signal (frequency-modulated)

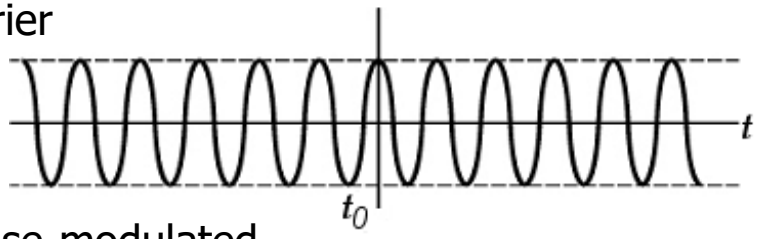


Analog (Continuous-Wave) Modulation

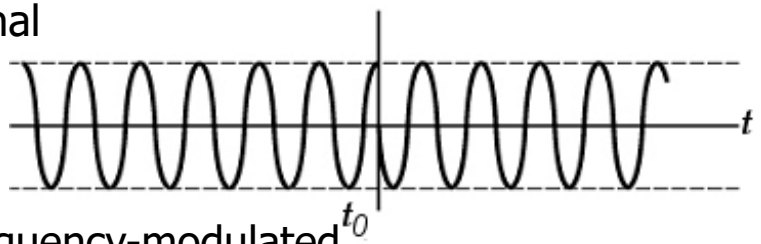
Message signal



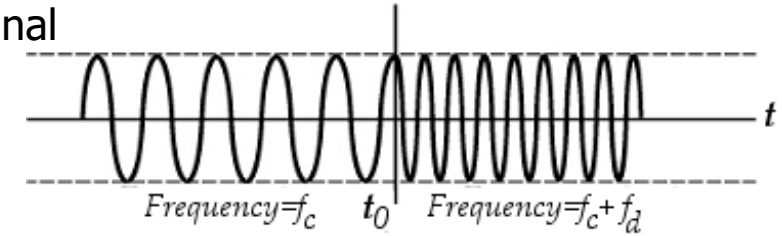
Unmodulated carrier

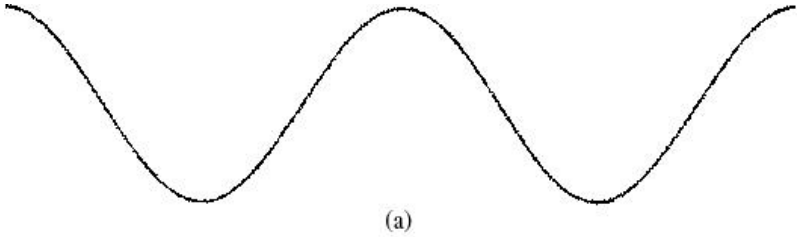


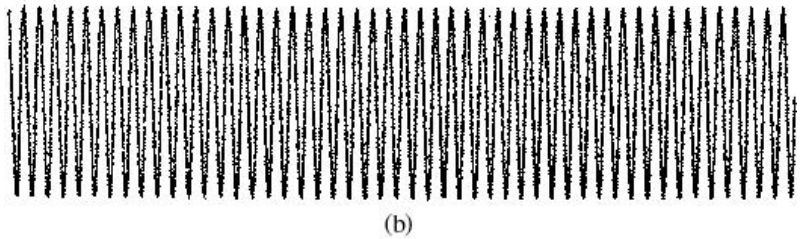
Phase-modulated signal

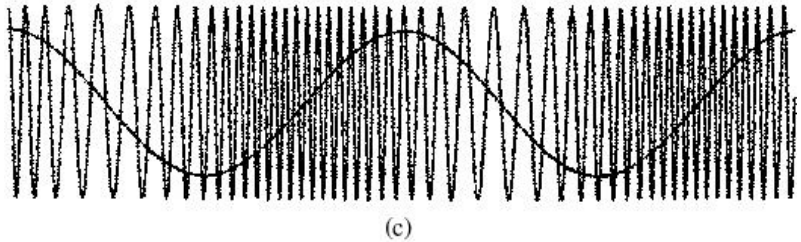


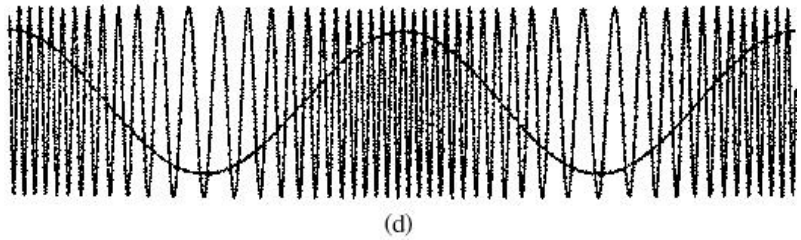
Frequency-modulated signal




(a)

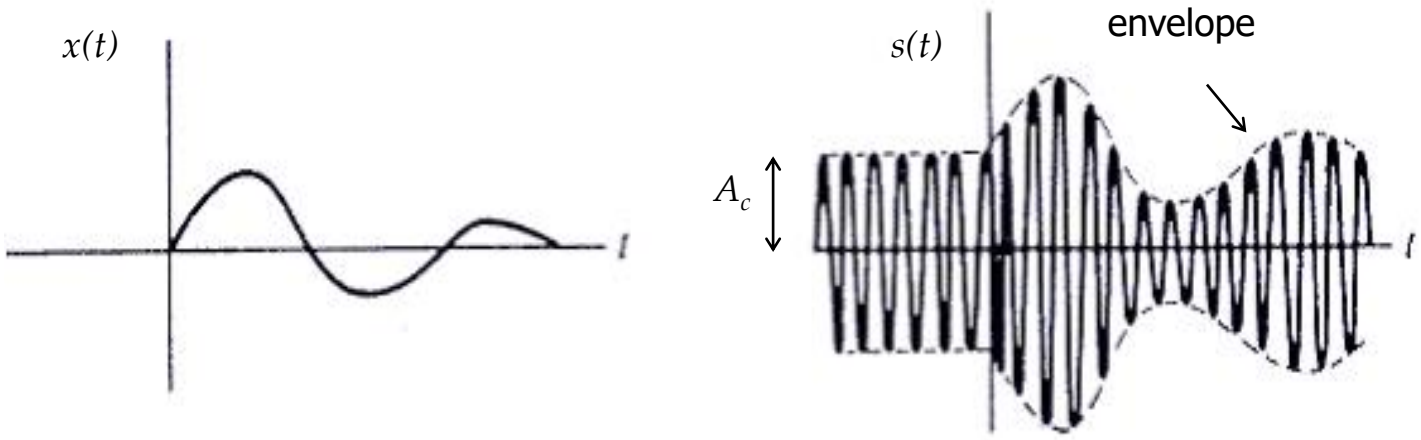

(b)


(c)


(d)

Amplitude Modulation

The envelope of the modulated carrier has the same shape as the message signal.



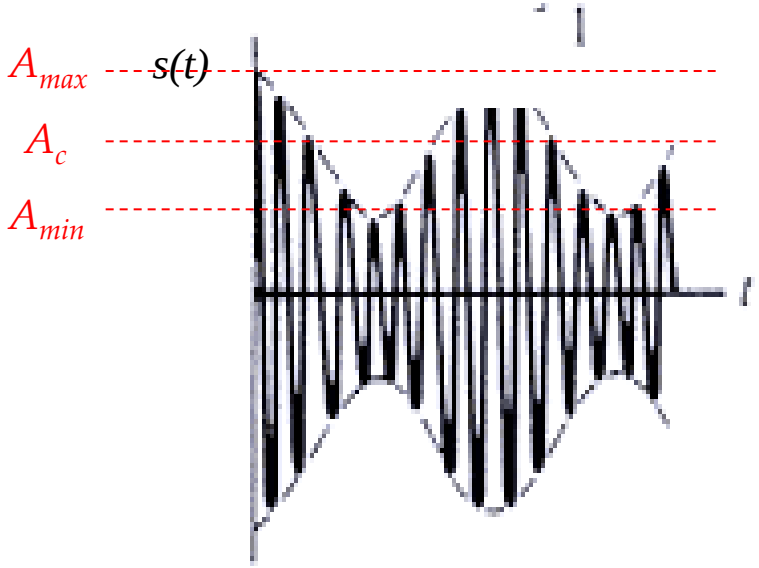
The amplitude of the carrier wave $[\ A_c \cos(2\pi f_c t)\ ]$ varies linearly with the baseband message signal $x(t)$.

Amplitude Modulation

Let $x(t)$ has amplitude a_m and its normalized form be $\tilde{x}(t)$ $\left(\text{i.e. } |\tilde{x}(t)| = 1\right)$

$$s(t) = \left[A_c + x(t)\right]\cos\omega_c t = \left[A_c + a_m \tilde{x}(t)\right]\cos\omega_c t$$

$$= A_c\left[1 + \frac{a_m}{A_c}\tilde{x}(t)\right]\cos\omega_c t$$



$$\left.\begin{aligned} A_{\max} &= A_c + a_m \\ A_{\min} &= A_c - a_m \end{aligned}\right\} \Rightarrow \begin{cases} A_c = \dfrac{A_{\max} + A_{\min}}{2} \\ a_m = \dfrac{A_{\max} - A_{\min}}{2} \end{cases}$$

$$\Rightarrow \frac{a_m}{A_c} = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}} = m_a$$

Standard form of an amplitude-modulated (AM) signal: $s(t) = A_c\left[1 + m_a \tilde{x}(t)\right]\cos\omega_c t$

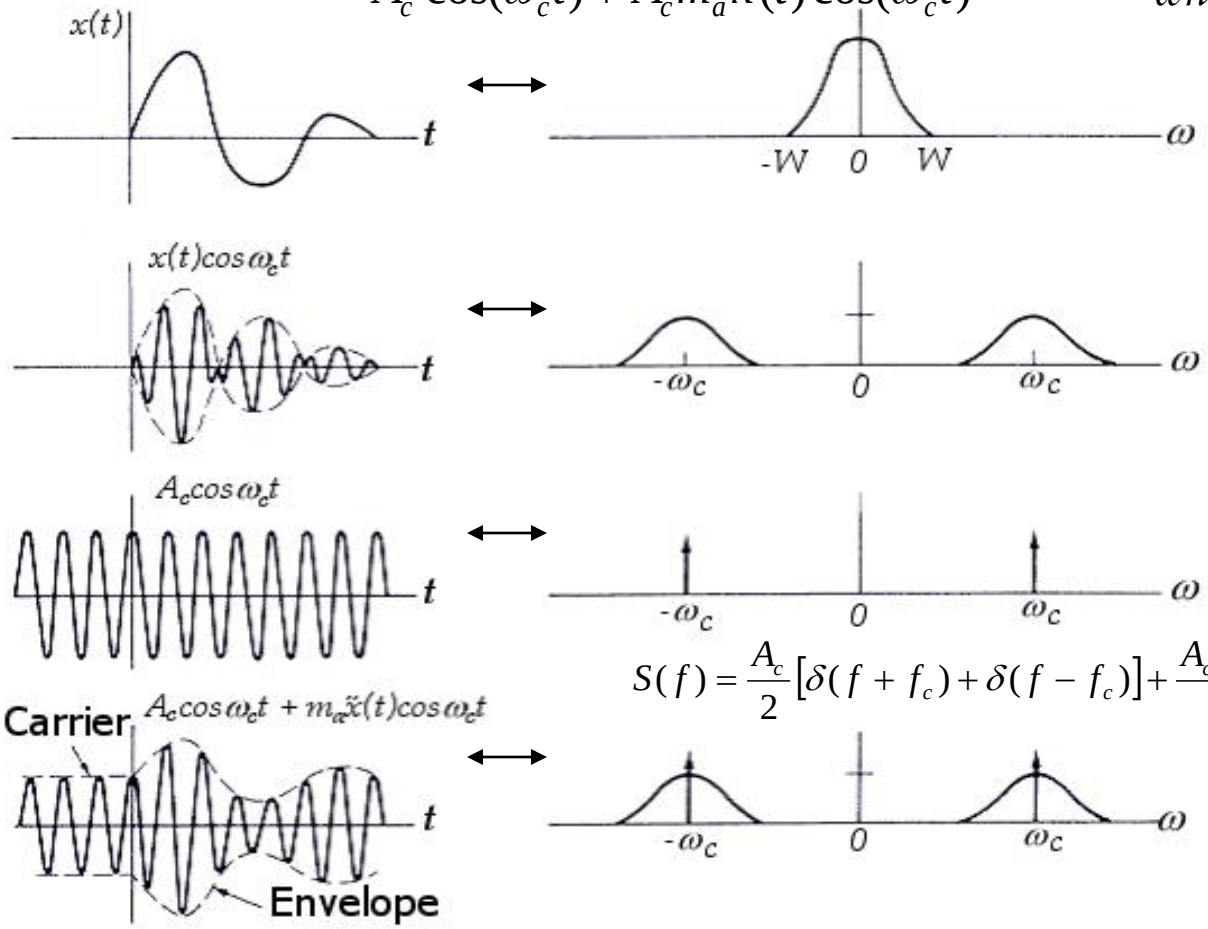
Modulation depth or modulation index: $m_a = \dfrac{a_m}{A_c}$

Amplitude Modulation: DSB-LC

AM signal: $s(t) = A_c\left[1 + m_a \tilde{x}(t)\right]\cos(2\pi\ f_c t)$

$= A_c\cos(2\pi f_c t) + A_c m_a \tilde{x}(t)\cos(2\pi f_c t)$

$= A_c\cos(\omega_c t) + A_c m_a \tilde{x}(t)\cos(\omega_c t)$ *where* $\omega_c = 2\pi f_c$



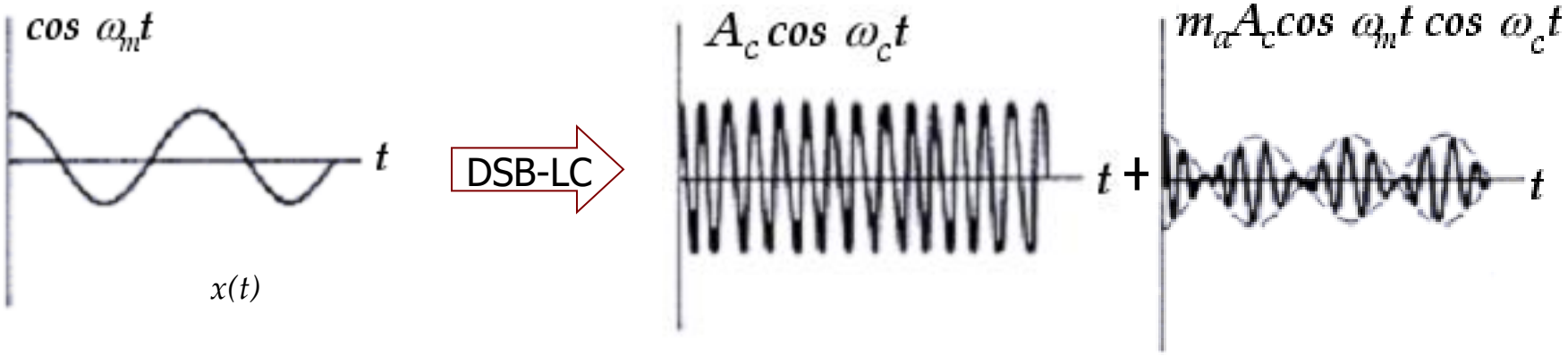
$$S(f) = \frac{A_c}{2}\left[\delta(f + f_c) + \delta(f - f_c)\right] + \frac{A_c m_a}{2}\left[X(f + f_c) + X(f - f_c)\right]$$

Double-sideband – large carrier (DSB-LC)

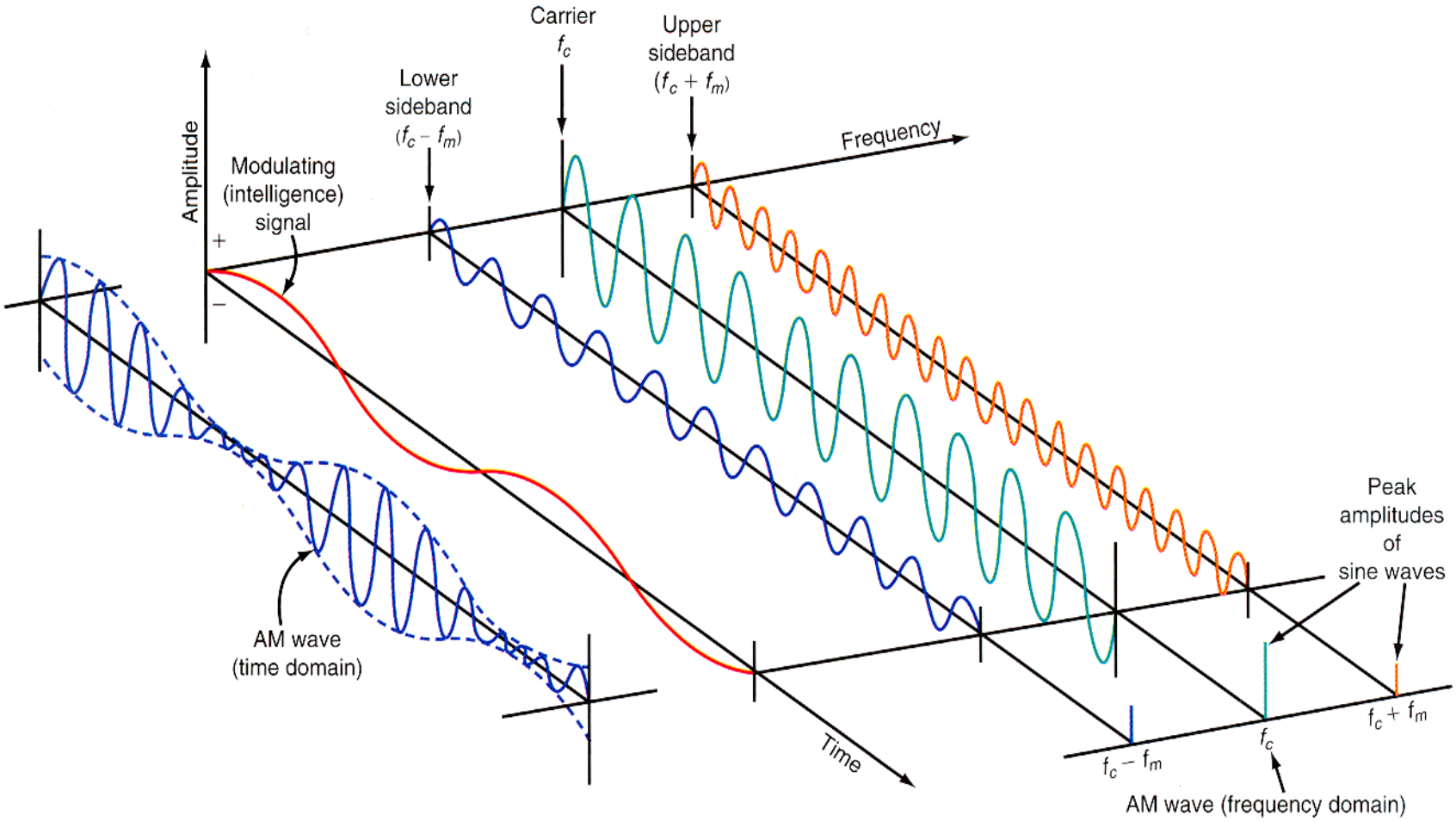
Amplitude Modulation

$$s(t) = A_c\left[1 + m_a \tilde{x}(t)\right]\cos\omega_c t$$

If $\tilde{x}(t) = \cos\omega_m t$, then $s(t) = A_c \cos\omega_c t + m_a A_c \cos\omega_m t \cos\omega_c t$

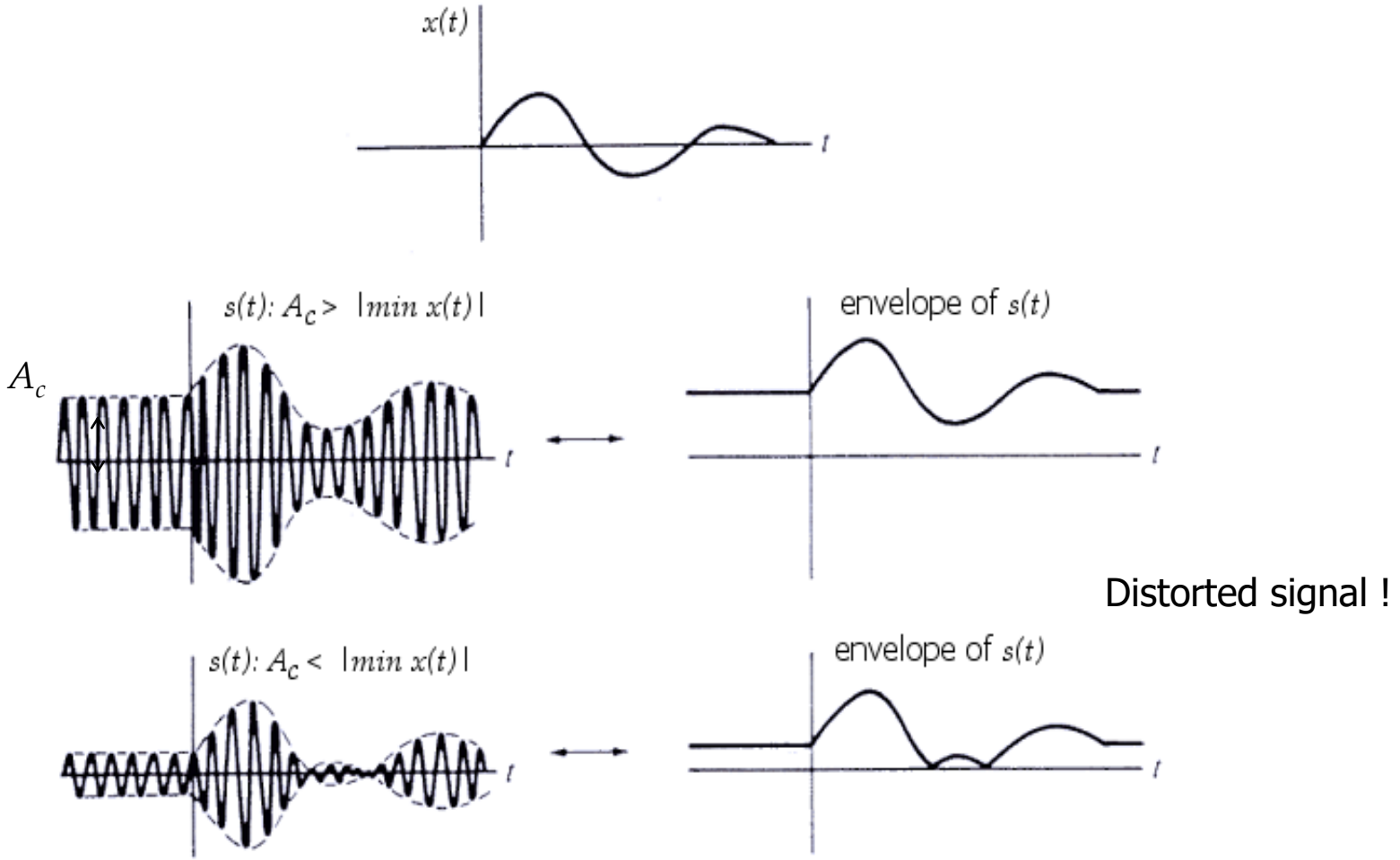


Amplitude Modulation



Amplitude Modulation: DSB-LC

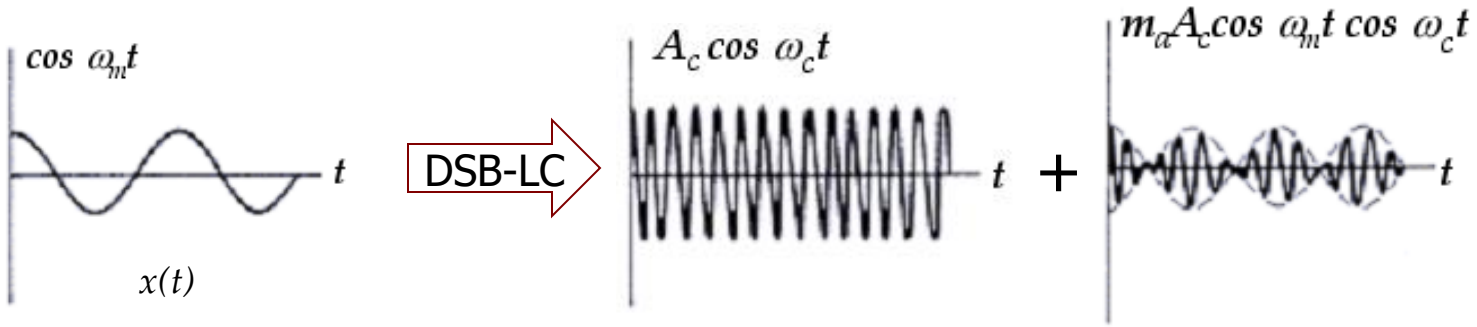
$$s(t) = A_c\left[1 + m_a \tilde{x}(t)\right]\cos(2\pi\ f_c t) = \left[A_c + x(t)\right]\cos(2\pi\ f_c t)$$



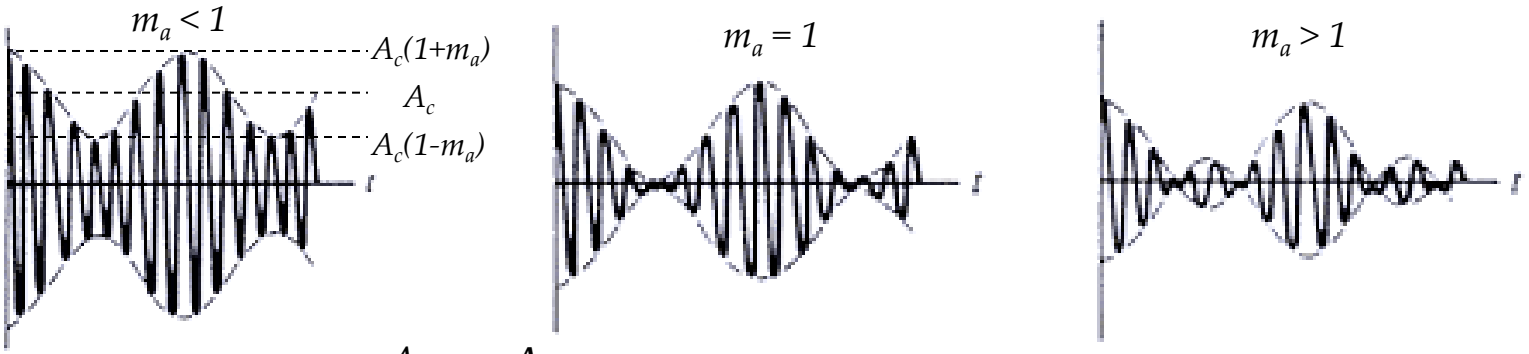
So, A_c has to be large enough or we have to control the modulation index m_a.

Amplitude Modulation: DSB-LC

$$s(t) = A_c\left[1 + m_a \tilde{x}(t)\right]\cos(2\pi\ f_c t)$$



Effect of modulation index m_a :



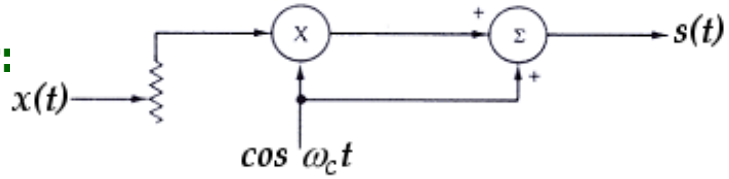
Modulation depth = $\dfrac{A_{max} - A_c}{A_c}$

For a sinusoidal message signal, $A_{max}=A_c(1+m_a)$, thus the modulation depth is m_a .

Amplitude Modulation: DSB-LC

Generation of DSB-LC signal: $s(t) = A_c\left[1 + m_a \tilde{x}(t)\right]\cos(2\pi\ f_c t)$

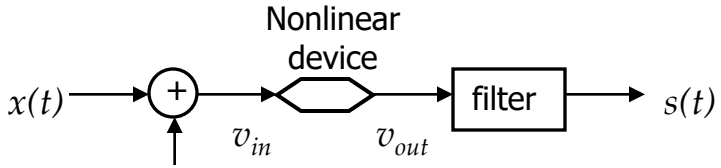
Product modulator:



Square-law modulator:

$$v_{out} = a_1 v_{in} + a_2 v_{in}^2$$

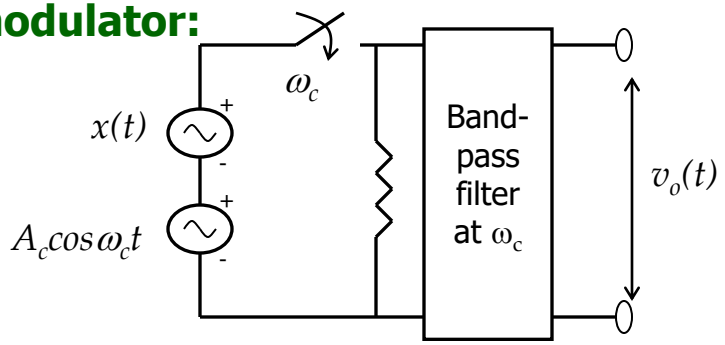
$$v_{in} = x(t) + \cos\omega_c t$$

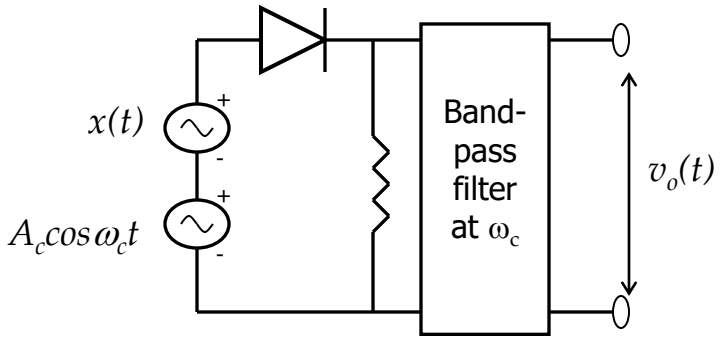


$$v_{out} = a_1 x(t) + a_2 x^2(t) + a_2 \cos^2 \omega_c t + \underbrace{a_1\left[1 + \frac{2a_2}{a_1} x(t)\right]\cos\omega_c t}_{s(t)}$$

where $A_c = a_1$ and $m_a = 2a_2/a_1$

Chopper/rectifier modulator:

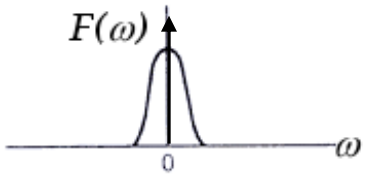




Amplitude Modulation: DSB-LC

Chopper/rectifier modulator:

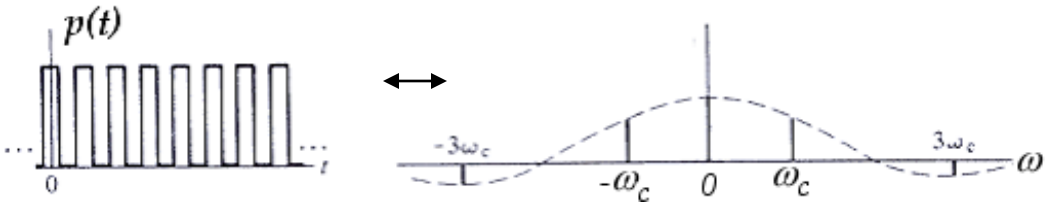
The chopper or rectifier can generate a periodic waveform whose fundamental frequency is ω_c rad/sec.



Let $f(t) = A_c + x(t)$

The periodic signal $p(t)$ can be represented as

$$p(t) = \sum_{n=-\infty}^{\infty} P_n e^{jn\omega_c t}$$

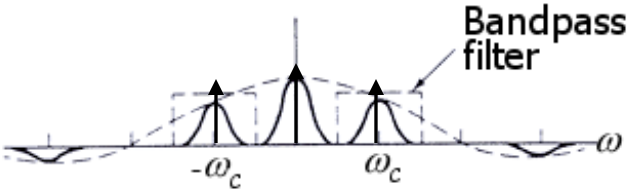


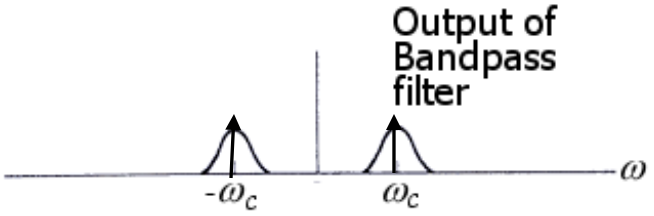
Consider $p(t)f(t)$:

$$f(t)p(t) = \sum_{n=-\infty}^{\infty} P_n f(t) e^{jn\omega_c t}.$$

Applying the frequency translation property of the Fourier transform, we get

$$\Im\{f(t)p(t)\} = \sum_{n=-\infty}^{\infty} P_n F(\omega - n\omega_c).$$



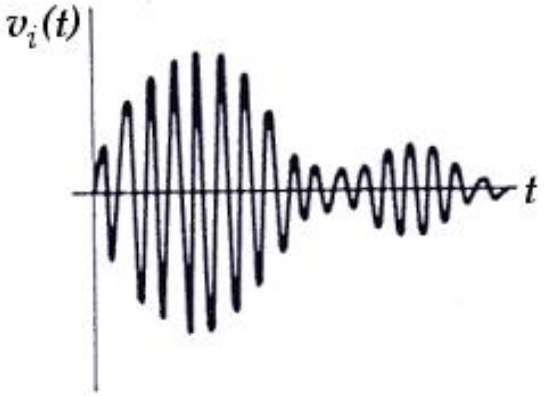


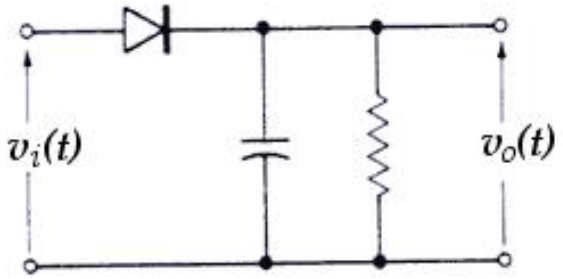
Amplitude Modulation: DSB-LC

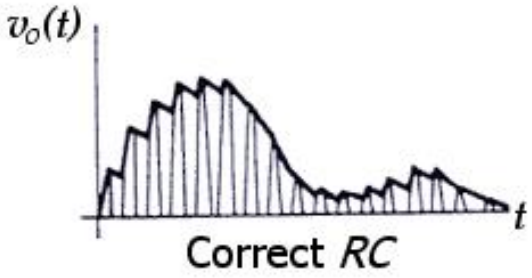
Demodulation of DSB-LC signal:

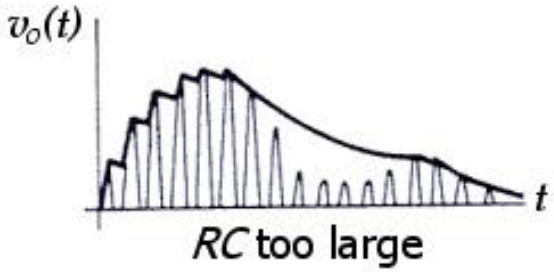
By envelope detector: the diode cuts off the negative part of the DSB-LC signal while RC acts as a lowpass filter to retrieve the envelope.

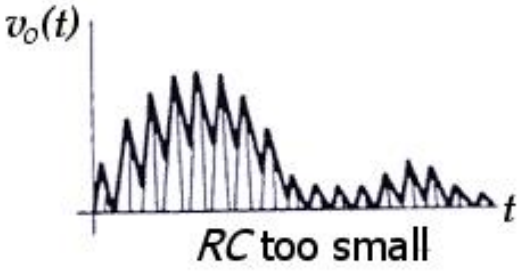
$1/W >> RC >> 1/f_c$ where W is the message signal bandwidth











Amplitude Modulation: DSB-LC

DSB-LC signal: $s(t) = A_c\left[1 + m_a \tilde{x}(t)\right]\cos(2\pi\ f_c t)$

Consider the **average power of** $s(t)$:

$$P_s = \overline{s^2(t)}$$

$$= \overline{A_c^2\left[1 + m_a \tilde{x}(t)\right]^2 \cos^2(\omega_c t)}$$

$$= A_c^2\left\{\overline{\left[1 + m_a^2 \tilde{x}^2(t) + 2m_a \tilde{x}(t)\right]\frac{1}{2}\left[1 + \cos(2\omega_c t)\right]}\right\}$$

$$= \frac{A_c^2}{2}\left\{\overline{\left[1 + m_a^2 \tilde{x}^2(t) + 2m_a \tilde{x}(t)\right]} + \overline{\left[1 + m_a^2 \tilde{x}^2(t) + 2m_a \tilde{x}(t)\right]\cos(2\omega_c t)}\right\}$$

$$\cong \frac{A_c^2}{2}\left[1 + m_a^2 \overline{\tilde{x}^2(t)}\right] \quad \text{if} \quad \overline{\tilde{x}(t)} = 0 \quad \text{and} \quad \tilde{x}(t) \text{ varies slowly with respect to } \cos(\omega_c t)$$

$$= \frac{A_c^2}{2}\left[1 + m_a^2 P_x\right] \quad \text{where} \quad P_x = \overline{\tilde{x}^2(t)}$$

$$= P_c + 2P_{sb}$$

where P_{sb} : average power per sideband

Thus $\quad P_c = \frac{1}{2}A_c^2 \ ; \ P_{sb} = \frac{1}{4}A_c^2 m_a^2 P_x = \frac{1}{2}m_a^2 P_x P_c$

For $\left|m_a \tilde{x}(t)\right| \le 1 \quad \Rightarrow \quad m_a^2 P_x \le 1 \quad \Rightarrow \quad P_{sb} \le \frac{1}{2}P_c \quad \Rightarrow \quad P_c = P_s - 2P_{sb} \ge \frac{1}{2}P_s$

At least **50%** of total transmitted power resides in the carrier term which conveys no information → **wasteful of power**

Amplitude Modulation: DSB-LC

Fraction of total transmitted power contained in the sidebands is:

$$\mu = \frac{2P_{sb}}{P_c + 2P_{sb}} = \frac{m_a^2 P_x P_c}{P_c + m_a^2 P_x P_c} = \frac{m_a^2 P_x}{1 + m_a^2 P_x}$$

If $x(t)$ is a single sinusoid, i.e. $\cos\omega_m t$, then $P_x = \overline{x^2(t)} = \frac{1}{2}$

Thus, $\mu = \frac{m_a^2}{2 + m_a^2}$ and is known as the ***transmission efficiency*** *of DSB-LC AM system.*

Example:

A given **AM** (DSB-LC) **broadcast station** transmits an average carrier power output of 40kW and uses a modulation index of 0.707 for sine-wave modulation.
Calculate (a) the total average power output; (b) the transmission efficiency; and
(c) the peak amplitude of the output if the antenna is represented by a 50-ohm resistive load.

Solution: (a) The total average power output is $P_s = P_c + 2P_{sb} = P_c(1 + m_a^2/2).$

For $m_a = 0.707$, $P_s = 40(1 + 1/4) = 50\ kW.$

(b) The transmission efficiency is $\mu = \frac{(0.707)^2}{2 + (0.707)^2} = \frac{0.5}{2.5} = 20\%.$

(c) Consider $P_c = \frac{A^2}{2R}, \Rightarrow A^2 = 2RP_c = 4\times10^6.$

The **peak amplitude** of the output is $(1 + m_a)A = 3414V.$

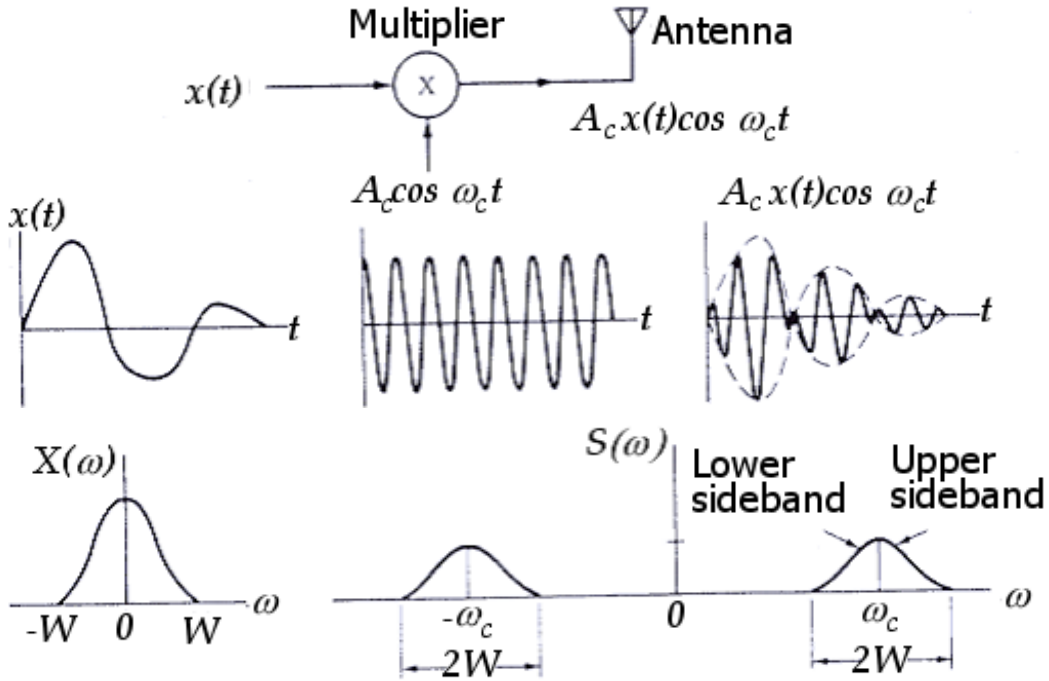
Amplitude Modulation: DSB-SC

The "wasted" carrier power in DSB-LC can be eliminated by setting $m_a=1$ and suppressing the carrier.

Thus the modulated signals becomes $s(t) = A_c \tilde{x}(t)\cos(2\pi\ f_c t)$

In this case, the carrier frequency component is suppressed, thus it is called ***double-sideband suppressed-carrier modulation (DSB-SC)*** .

Its spectral density is: $S(f) = \frac{A_c}{2}[X(f+f_c)+X(f-f_c)]$



Average Power of the modulated signal:

$$P_s = \overline{s^2(t)}$$
$$= \overline{A_c^2 \tilde{x}^2(t)\cos^2(\omega_c t)}$$
$$= \frac{1}{2}A_c^2 P_x \quad where \quad P_x = \overline{\tilde{x}^2(t)}$$
$$= 2P_{sb}$$
$$\therefore P_{sb} = \frac{1}{4}A_c^2 P_x$$

Amplitude Modulation: DSB-SC

A signal spectrum can be translated an amount $\pm\,\omega_c$ rad/sec in frequency by multiplying the signal with any periodic waveform whose fundamental frequency is ω_c rad/sec.

The periodic signal $p(t)$ can be represented as

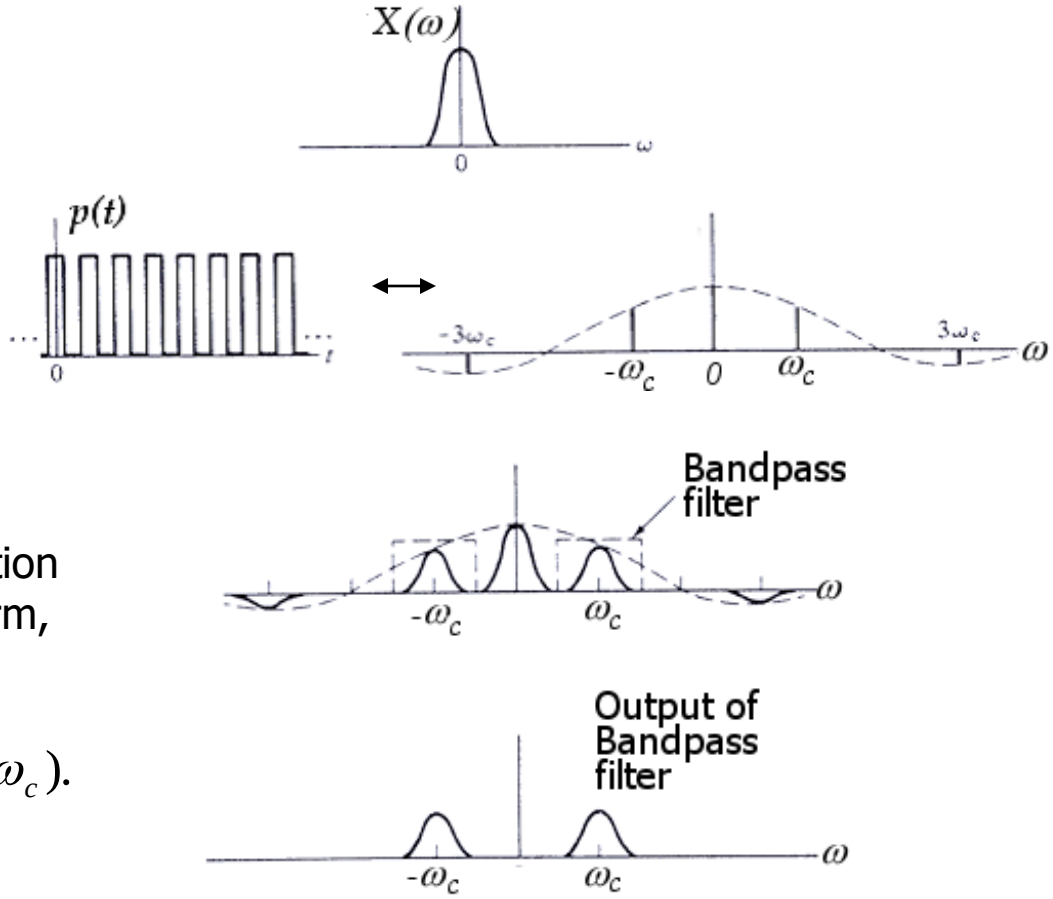
$$p(t) = \sum_{n=-\infty}^{\infty} P_n e^{jn\omega_c t}$$

Consider $p(t)x(t)$:

$$x(t)\,p(t) = \sum_{n=-\infty}^{\infty} P_n x(t) e^{jn\omega_c t}.$$

Applying the frequency translation property of the Fourier transform, we get

$$\Im\{x(t)\,p(t)\} = \sum_{n=-\infty}^{\infty} P_n X(\omega - n\omega_c).$$



Amplitude Modulation: DSB-SC

Example:

A periodic signal consists of the exponentially decreasing waveform e^{-at}, $0 \le t < T$, repeated every T seconds. A given signal $f(t)$ is multiplied by this periodic signal. Determine an expression describing the spectrum and the time waveform of the resulting amplitude-modulated signal if all components except those centered at $\pm\omega_c$, $\omega_c = 2\pi/T$, are discarded.

Solution: The Fourier series for the given periodic signal can be written as

$$p(t) = \sum_{n=-\infty}^{\infty} P_n e^{jn\omega_c t}, \quad \text{where} \quad P_n = \frac{1}{T}\int_0^T e^{-at} e^{-jn\omega_c t}\,dt = \frac{1}{T}\frac{1-e^{-aT}}{a+jn\omega_c}.$$

The spectrum of the product $p(t)f(t)$ after the BPF is

$$\frac{1}{T}(1-e^{-aT})\left[\frac{1}{a+j\omega_c}F(\omega-\omega_c)+\frac{1}{a-j\omega_c}F(\omega+\omega_c)\right].$$

The corresponding terms in time domain are

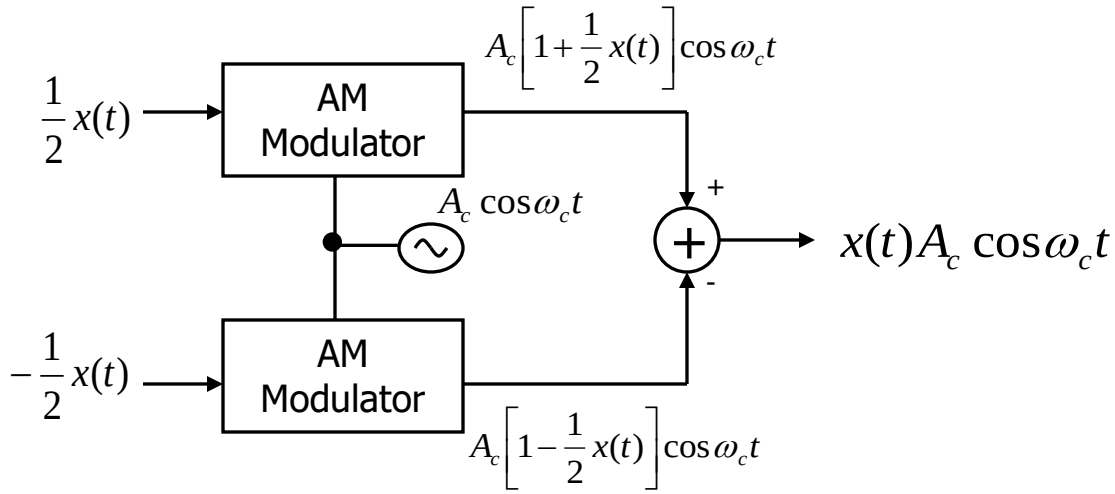
$$\frac{1}{T}(1-e^{-aT})\left[\frac{1}{a+j\omega_c}f(t)e^{j\omega_c t}+\frac{1}{a-j\omega_c}f(t)e^{-j\omega_c t}\right].$$

Combining yields the time waveform

$$\frac{2}{T}\frac{1-e^{-aT}}{\sqrt{a^2+\omega_c{}^2}}f(t)\cos(\omega_c t+\theta_0), \quad \text{where} \quad \theta_0 = \tan^{-1}(-\omega_c/a), \quad \omega_c = 2\pi/T.$$

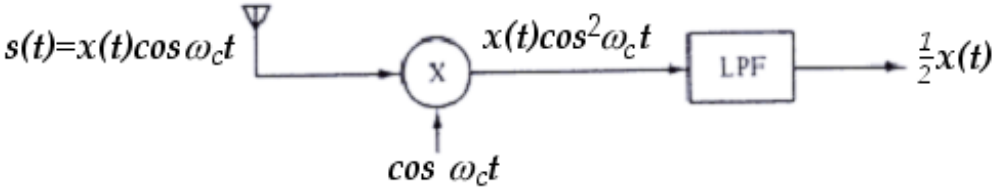
Amplitude Modulation: DSB-SC

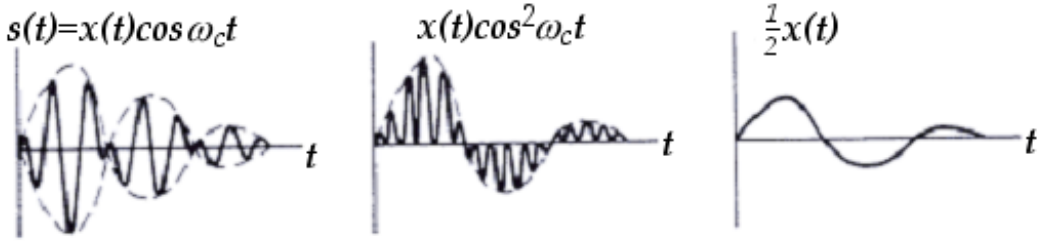
Generation of DSB-SC signal by balanced modulator:

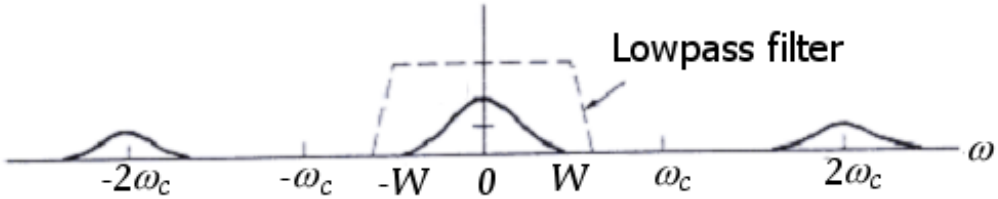


Amplitude Modulation: DSB-SC

Demodulation of DSB-SC signal:







Assuming that the transmitted signal is

$$s(t) = x(t)\cos\omega_c t$$

To demodulate the signal, we have

$$s(t)\cos\omega_c t = x(t)\cos^2\omega_c t$$
$$= \tfrac{1}{2}x(t) + \tfrac{1}{2}x(t)\cos 2\omega_c t$$

Taking the Fourier transform of both sides, we get

$$\Im\{s(t)\cos\omega_c t\} = \tfrac{1}{2}X(\omega)$$
$$+ \tfrac{1}{4}X(\omega + 2\omega_c)$$
$$+ \tfrac{1}{4}X(\omega - 2\omega_c)$$

Amplitude Modulation: DSB-SC

Consider a small frequency error, $\Delta\omega$, and a phase error, θ_0, are introduced in the locally generated carrier signal at the receiver. The signal at the receiver becomes

$$\begin{aligned} s(t)\cos[(\omega_c+\Delta\omega)t+\theta_0] &= x(t)\cos\omega_c t\cos[(\omega_c+\Delta\omega)t+\theta_0] \\ &= \tfrac{1}{2}x(t)\cos[(\Delta\omega)t+\theta_0] \\ &\quad +\tfrac{1}{2}x(t)\cos[(2\omega_c+\Delta\omega)t+\theta_0]. \end{aligned}$$

After passing via the low-pass filter, the output is

$$e_o(t)=\tfrac{1}{2}x(t)\cos[(\Delta\omega)t+\theta_0].$$

Phase error and frequency error results in undesirable distortion. In some cases, they vary randomly, resulting in unacceptable performance.

Remedy:

Using a synchronized oscillator to recover the original signal $x(t)$ from the modulated signal $s(t)$. (Synchronous detection, or coherent detection)

Amplitude Modulation: DSB-SC

The original signal $x(t)$ can be recovered from the modulated signal $s(t)$ by multiplying $s(t)$ by $cos\omega_c t$ (i.e. synchronous detection).

The same circuits as those used for modulation can be used for demodulation with the following minor differences.

1. Since the desired output spectrum is centered about $\omega=0$ and therefore a low-pass filter is needed at the output.

2. The oscillator in the demodulator must be synchronized to the oscillator in the modulator to achieve proper demodulation.

   This is usually accomplished by either a direct connection if the modulator and demodulator are in close proximity
   or
   by supplying a sinusoid displaced in frequency but related to the modulator-oscillator frequency. The sinusoid is called a "pilot carrier".

Amplitude Modulation: DSB-SC

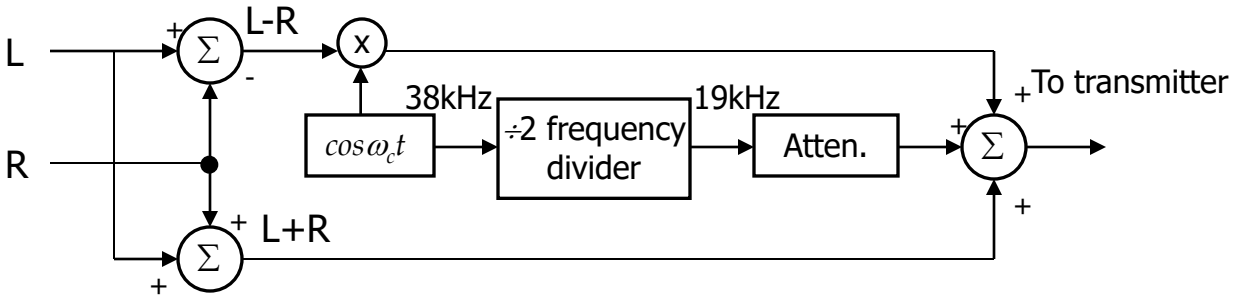
Pilot Carrier Systems

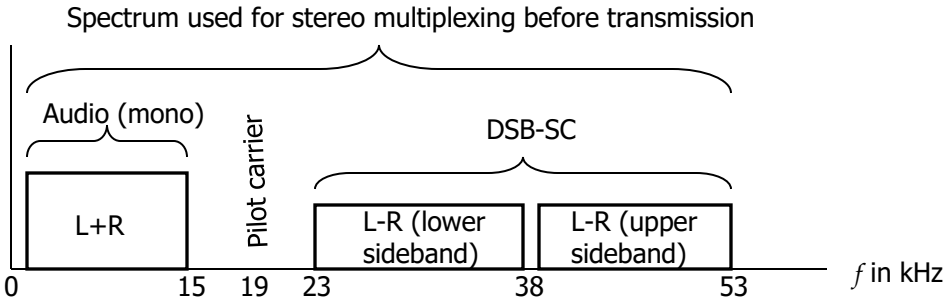
It is a common method used in DSB-SC modulation to maintain synchronization between modulator and demodulator.

In this case, a sinusoidal tone whose frequency and phase are related to the carrier frequency is generated and is sent outside the pass-band of the modulated signal so it will not alter the frequency response capability of the system.

A tuned circuit in the receiver detects the tone, translate it to the proper frequency, and uses it to correctly demodulate the DSB-SC signal.

e.g. **Stereo-multiplex system**

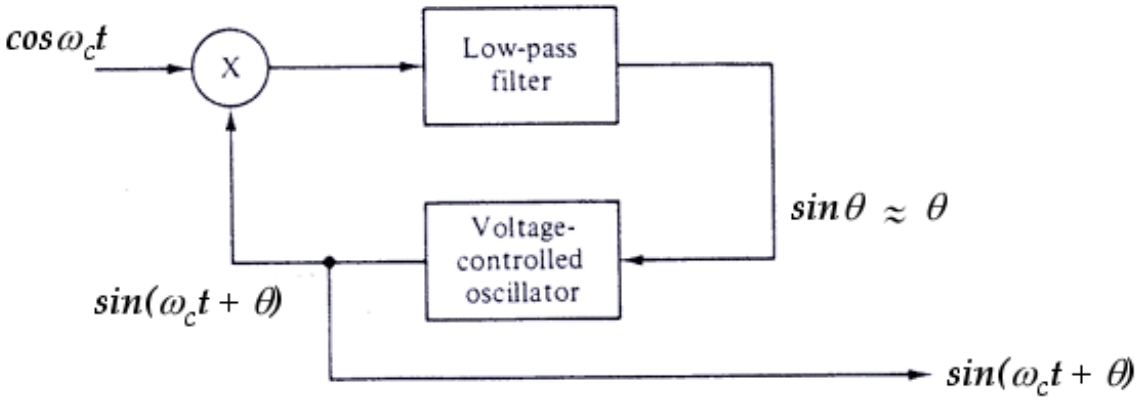




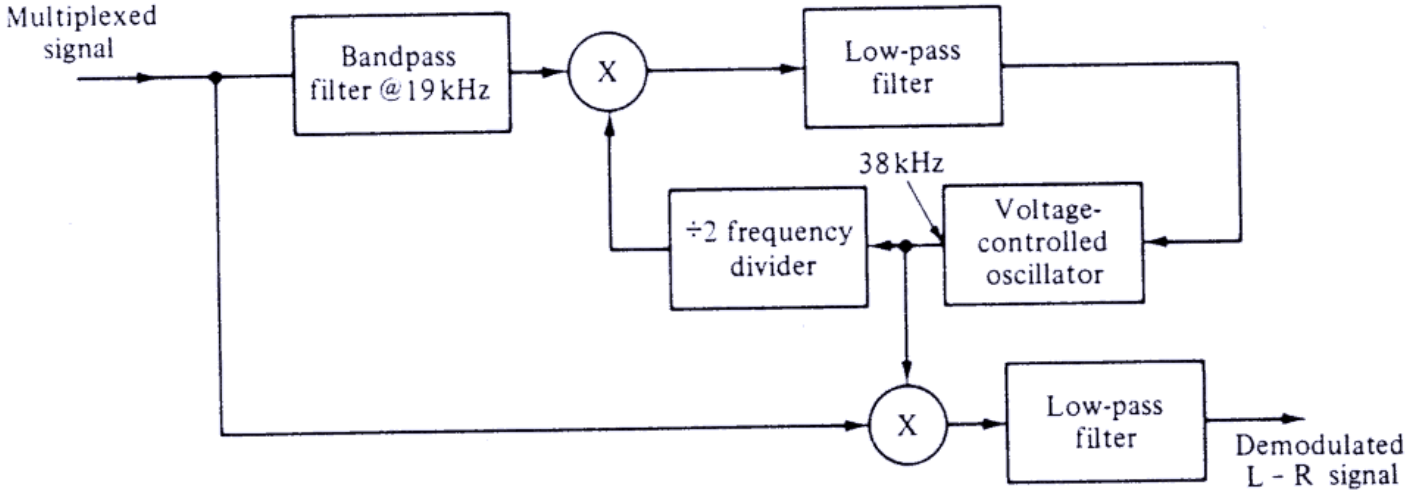
Amplitude Modulation: DSB-SC

Phase-Locked Loop (PLL)

In pilot tone system, phase-locked loop is used to synchronize one sinusoidal to another.



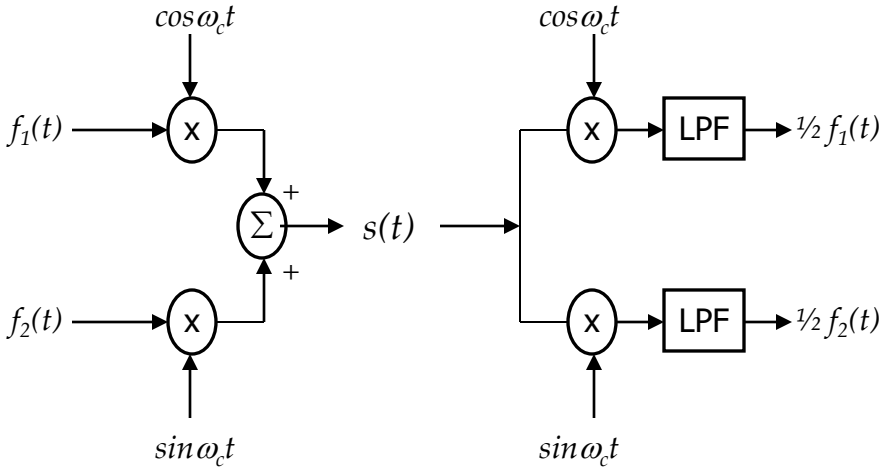
A simplified phase-locked loop stereo demodulator.



Amplitude Modulation: QAM

Quadrature Multiplexing

Using the orthogonality of sines and cosines , it is possible to transmit and receive two different signals simultaneously on the same frequency.



$$s(t) = f_1(t)\cos\omega_c t + f_2(t)\sin\omega_c t$$

$$s(t)\cos\omega_c t = f_1(t)\cos^2\omega_c t + f_2(t)\sin\omega_c\cos\omega_c t$$
$$= \tfrac{1}{2} f_1(t) + \tfrac{1}{2} f_1(t)\cos 2\omega_c t + \tfrac{1}{2} f_2(t)\sin 2\omega_c t$$

$$s(t)\sin\omega_c t = f_1(t)\cos\omega_c t\sin\omega_c t + f_2(t)\sin^2\omega_c t$$
$$= \tfrac{1}{2} f_1(t)\sin 2\omega_c t + \tfrac{1}{2} f_2(t) - \tfrac{1}{2} f_2(t)\cos 2\omega_c t$$

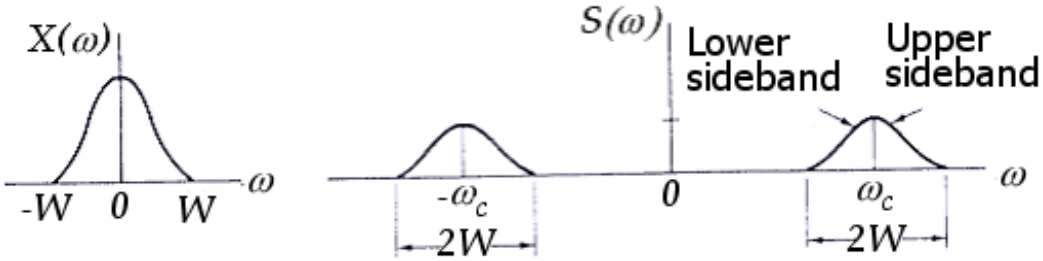
In the low-pass filter, all terms at $2\omega_c$ are attenuated, yielding

$$e_1(t) = \tfrac{1}{2} f_1(t),$$
$$e_2(t) = \tfrac{1}{2} f_2(t).$$

Thus, each signal can be recovered by synchronous detection of the received signal using carriers of the same frequency but in phase quadrature.

Amplitude Modulation: SSB

DSB modulation:



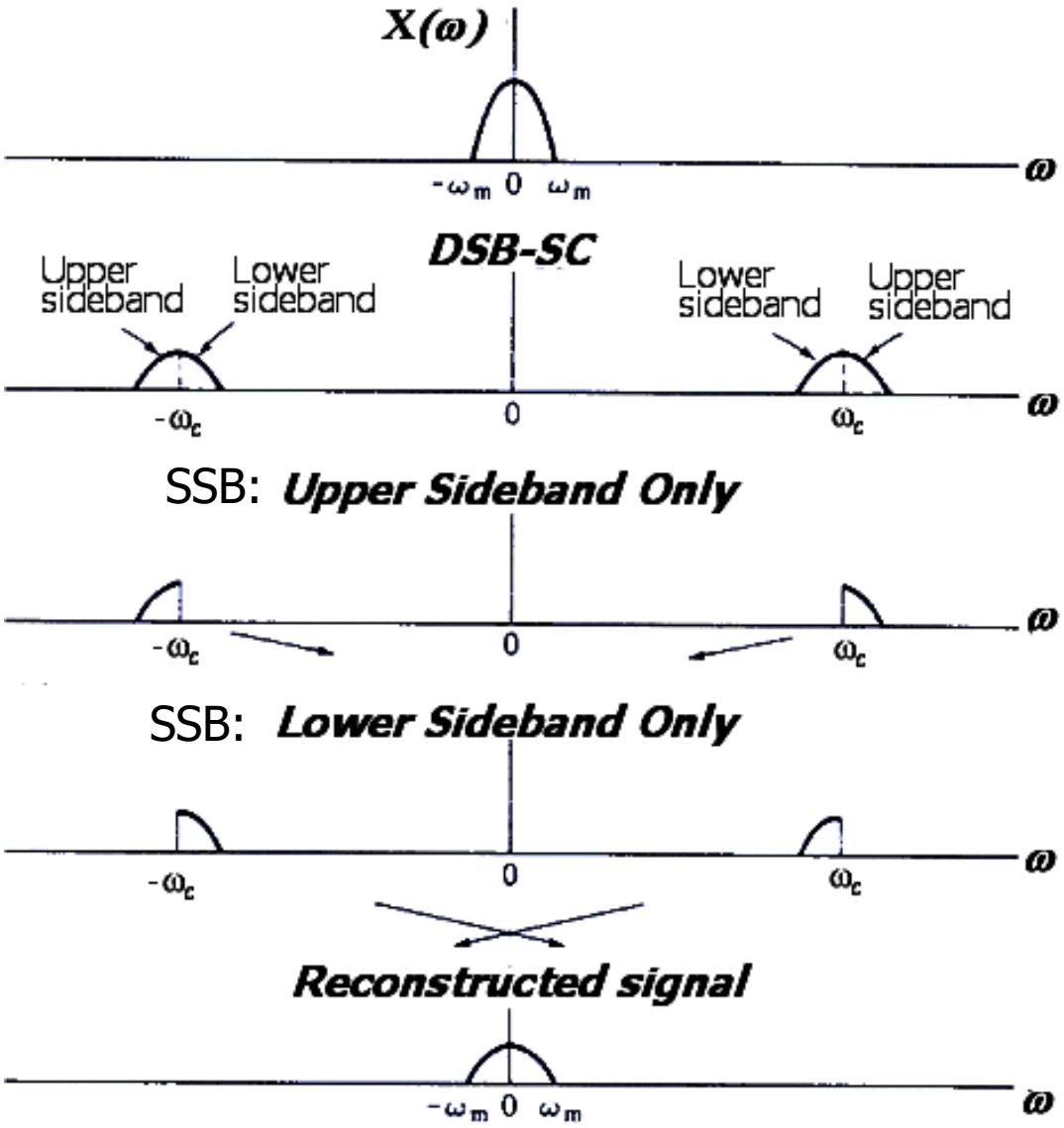
By mixing with a sinusoidal carrier at ω_c rad/sec, half of this spectral density is translated up in frequency and centered about ω_c and half is translated down to (- ω_c).

➔ **Doubling of the bandwidth** of a given signal

Since each half contains all the information about the signal, the original signal can be recovered again from either the upper or lower pair of sidebands by an appropriate frequency translation. So **only the upper or the lower pair of sidebands is required to transmit**.

Such kind of modulation is called ***single-sideband (SSB) modulation***. It is efficient because it requires no more bandwidth than that of the original signal and only half of the corresponding DSB signal.

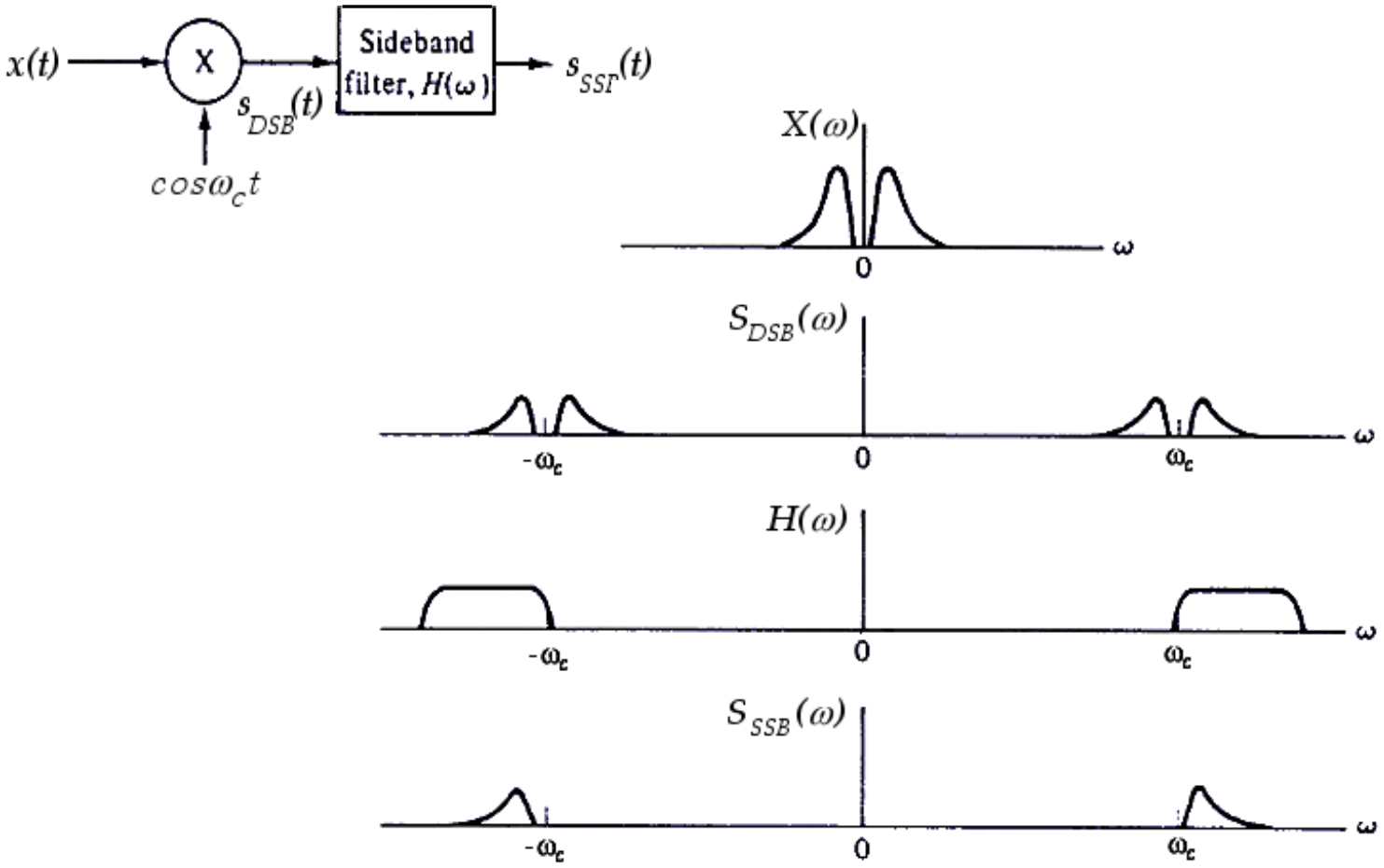
Amplitude Modulation: SSB



Amplitude Modulation: SSB

Generation of SSB Signals

-first generate a DSB signal, then suppress one of the sidebands by filtering.



Amplitude Modulation: SSB

Generation of SSB: Phase-Shift Method

Consider a modulating signal $x(t)$ where $x(t) = \cos\omega_m t$, for example.

Let $\hat{x}(t)$ be a $\pi/2$ phase-shifted version of $x(t)$, i.e. $\hat{x}(t) = \cos(\omega_m t - \pi/2) = \sin\omega_m t$

Their Fourier transforms: $X(w) = \mathcal{F}\{x(t)\} = \mathcal{F}\{\cos w_m t\} = p[d(w - w_m) + d(w + w_m)]$

$\hat{X}(w) = \mathcal{F}\{\hat{x}(t)\} = \mathcal{F}\{\sin w_m t\} = -jp[d(w - w_m) - d(w + w_m)]$

Consider:

$$\Im\{x(t)\cos\omega_c t\} = \frac{1}{2}[X(\omega+\omega_c) + X(\omega-\omega_c)]$$

$$= \frac{\pi}{2}[\delta(\omega+\omega_c-\omega_m) + \delta(\omega+\omega_c+\omega_m) + \delta(\omega-\omega_c-\omega_m) + \delta(\omega-\omega_c+\omega_m)]$$

$$\Im\{\hat{x}(t)\sin\omega_c t\} = \frac{j}{2}[\hat{X}(\omega+\omega_c) - \hat{X}(\omega-\omega_c)]$$

$$= \frac{\pi}{2}[\delta(\omega+\omega_c-\omega_m) - \delta(\omega+\omega_c+\omega_m) - \delta(\omega-\omega_c-\omega_m) + \delta(\omega-\omega_c+\omega_m)]$$

SSB upper sideband:

$$\Im\{s_{SSB+}(t)\} = \pi[\delta(\omega+\omega_c+\omega_m) + \delta(\omega-\omega_c-\omega_m)]$$

$$= \Im\{x(t)\cos\omega_c t\} - \Im\{\hat{x}(t)\sin\omega_c t\} \quad \Rightarrow \quad s_{SSB+}(t) = x(t)\cos\omega_c t - \hat{x}(t)\sin\omega_c t$$

SSB lower sideband:

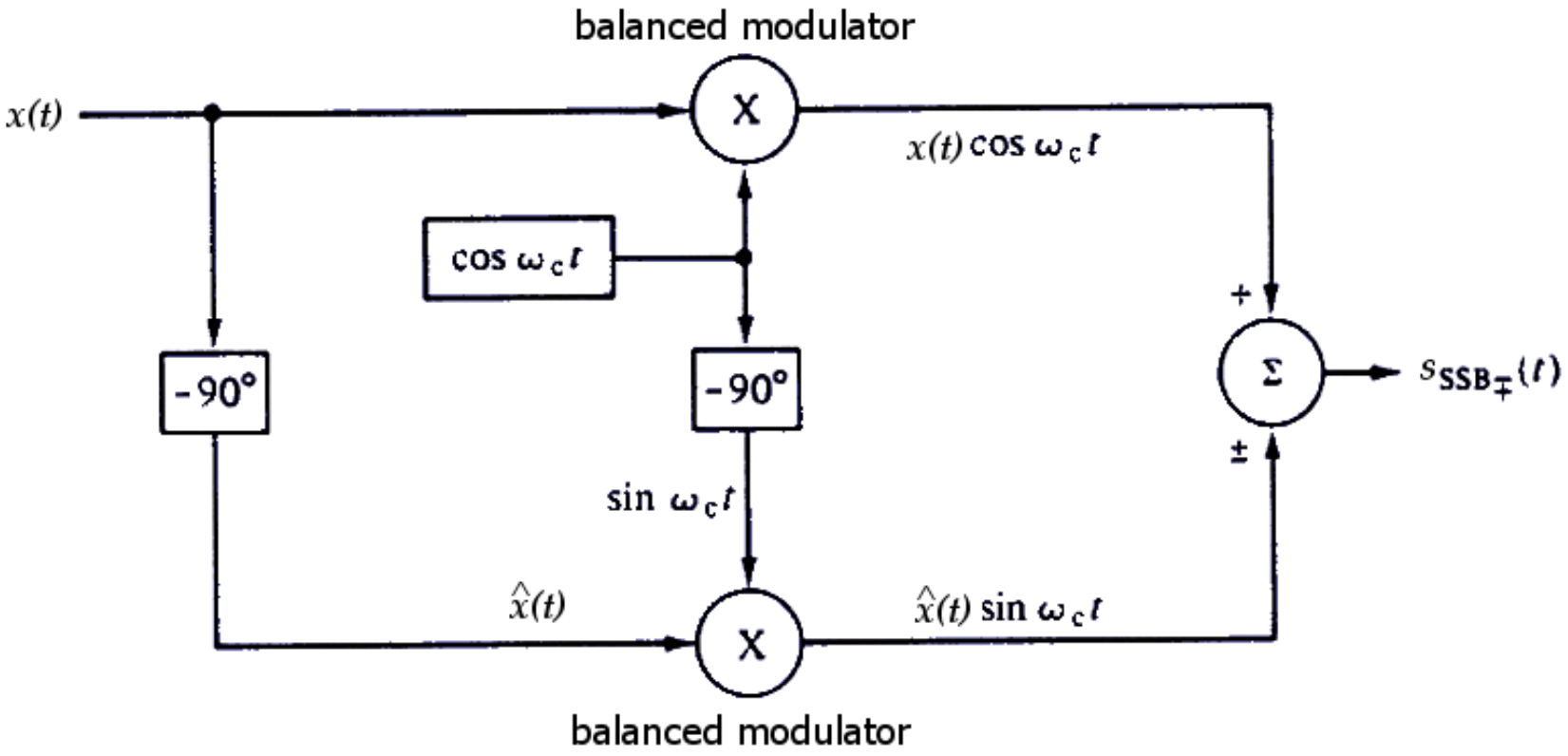
$$\Im\{s_{SSB-}(t)\} = \pi[\delta(\omega+\omega_c-\omega_m) + \delta(\omega-\omega_c+\omega_m)]$$

$$= \Im\{x(t)\cos\omega_c t\} + \Im\{\hat{x}(t)\sin\omega_c t\} \quad \Rightarrow \quad s_{SSB-}(t) = x(t)\cos\omega_c t + \hat{x}(t)\sin\omega_c t$$

Amplitude Modulation: SSB

Generation of SSB Signals : Phase-Shift Method

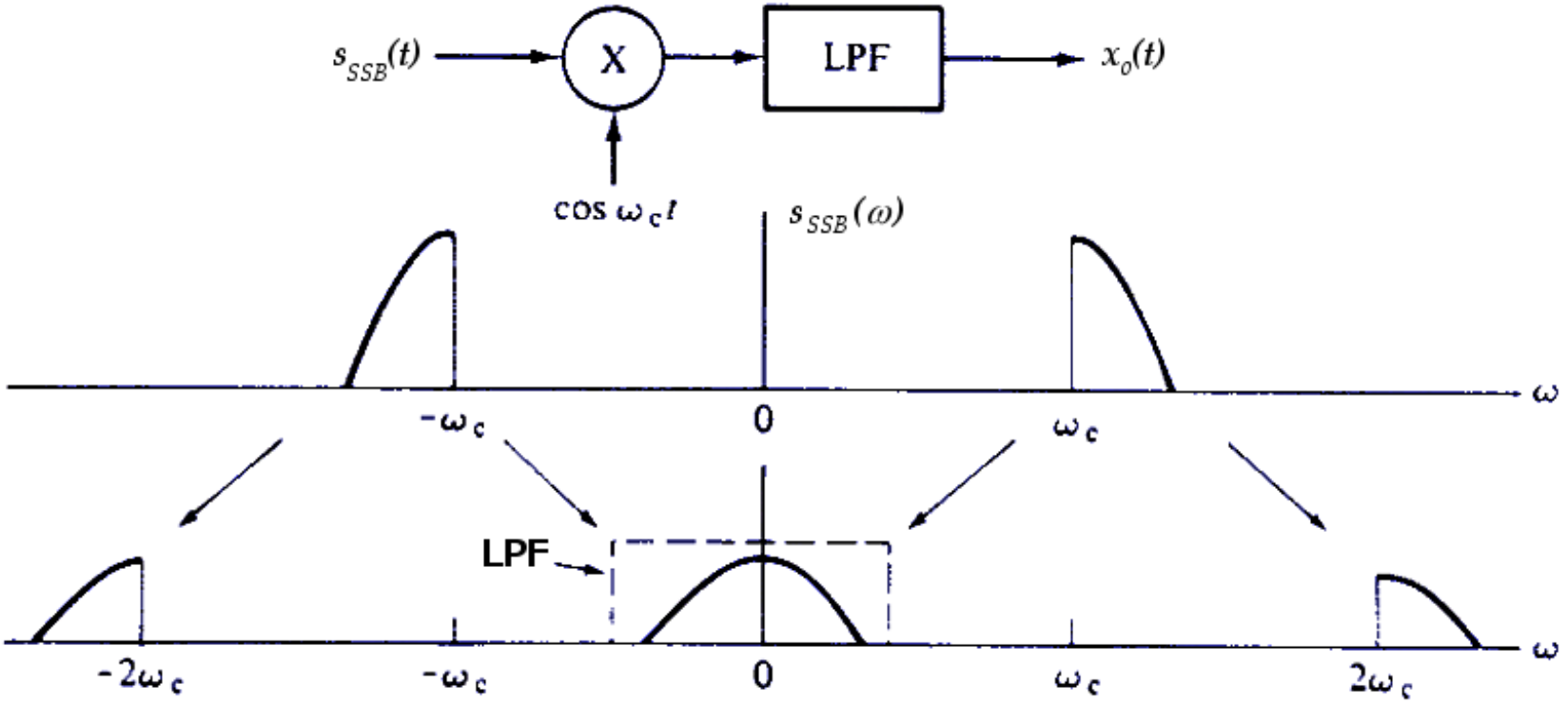
$$s_{SSB\mp}(t) = x(t)\cos\omega_c t \pm \hat{x}(t)\sin\omega_c t$$



Amplitude Modulation: SSB

Demodulation of SSB Signals

The synchronous detector will properly demodulate SSB-SC signals



Amplitude Modulation: SSB

Given that the incoming SSB-SC signal is $s_{SSB\mp}(t) = x(t)\cos\omega_c t \pm \hat{x}(t)\sin\omega_c t$

Let the locally generated carrier signal be $c(t) = \cos[(\omega_c + \Delta\omega)t + \theta]$,

where ($\Delta\omega$) is the frequency error and θ is the phase error.

$$\begin{aligned} s_{SSB\mp}(t)c(t) &= [x(t)\cos\omega_c t \pm \hat{x}(t)\sin\omega_c t]\cos[(\omega_c + \Delta\omega)t + \theta] \\ &= \tfrac{1}{2}x(t)\{\cos[(\Delta\omega)t + \theta] + \cos[(2\omega_c + \Delta\omega)t + \theta]\} \\ &\mp \tfrac{1}{2}\hat{x}(t)\{\sin[(\Delta\omega)t + \theta] - \sin[(2\omega_c + \Delta\omega)t + \theta]\} \end{aligned}$$

After passing through a low-pass filter, the output $x_o(t)$ becomes

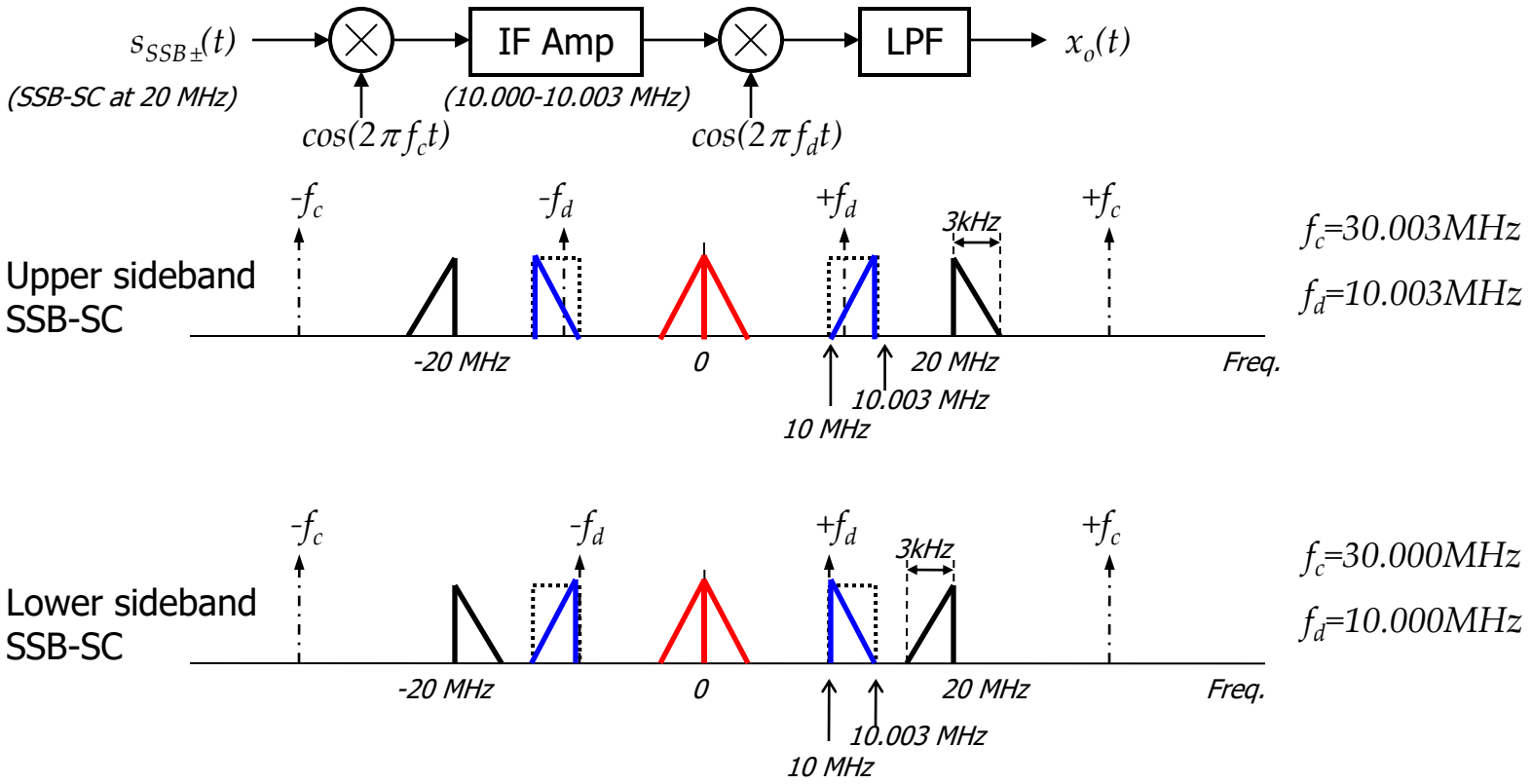
$$x_o(t) = \tfrac{1}{2}x(t)\cos[(\Delta\omega)t + \theta] \mp \tfrac{1}{2}\hat{x}(t)\sin[(\Delta\omega)t + \theta]$$ **Distorted !**

If $\Delta\omega = 0$ and $\theta = 0$, then $x_o(t) = \tfrac{1}{2}x(t)$

Amplitude Modulation: SSB

Example:

When an SSB (upper/lower-sideband) is received and fed into the following demodulator:



Amplitude Modulation: SSB

Single Sideband-Large Carrier (SSB-LC) Signals

An expression for an SSB-LC signal is

$$s(t) = A_c \cos\omega_c t + x(t)\cos\omega_c t \mp \hat{x}(t)\sin\omega_c t$$

The original signal $x(t)$ can always be recovered from $s(t)$ using synchronous detection.

If the carrier is large, however, envelope detection can also be used.

The envelope can be written as

$$r(t) = \sqrt{[A_c + x(t)]^2 + [\hat{x}(t)]^2} = A_c\sqrt{1 + \frac{2x(t)}{A_c} + \frac{x^2(t)}{A_c^2} + \frac{\hat{x}^2(t)}{A_c^2}}$$

If the carrier is much larger than the SSB-SC envelope, we have

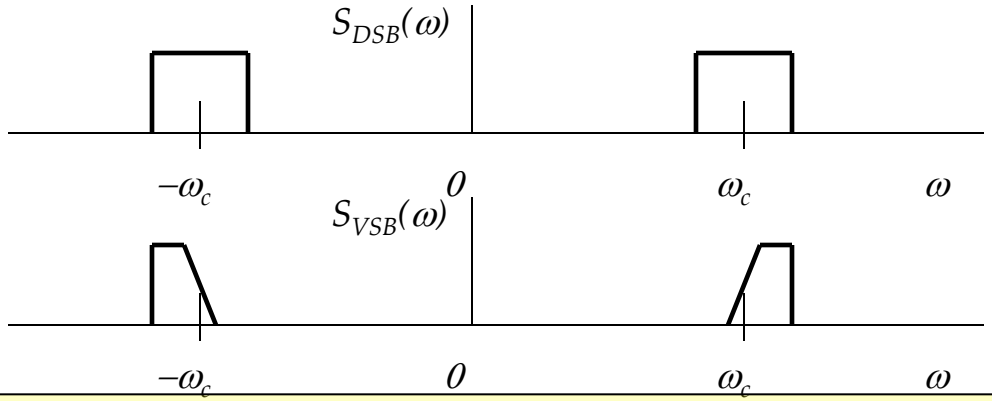
$$r(t) \approx A_c\sqrt{1 + \frac{2x(t)}{A_c}} \approx A_c\left[1 + \frac{x(t)}{A_c}\right] = A_c + x(t).$$

Thus after discarding the dc term introduced by carrier, the SSB-LC signal can then be demodulated correctly using an envelope detector.

Amplitude Modulation: VSB

Vestigial sideband (VSB) modulation is a compromise between DSB and SSB.

In VSB modulation, one passband is passed almost completely whereas only a residual portion of the other sideband is retained in such a way that the demodulation process can still reproduce the original signal.



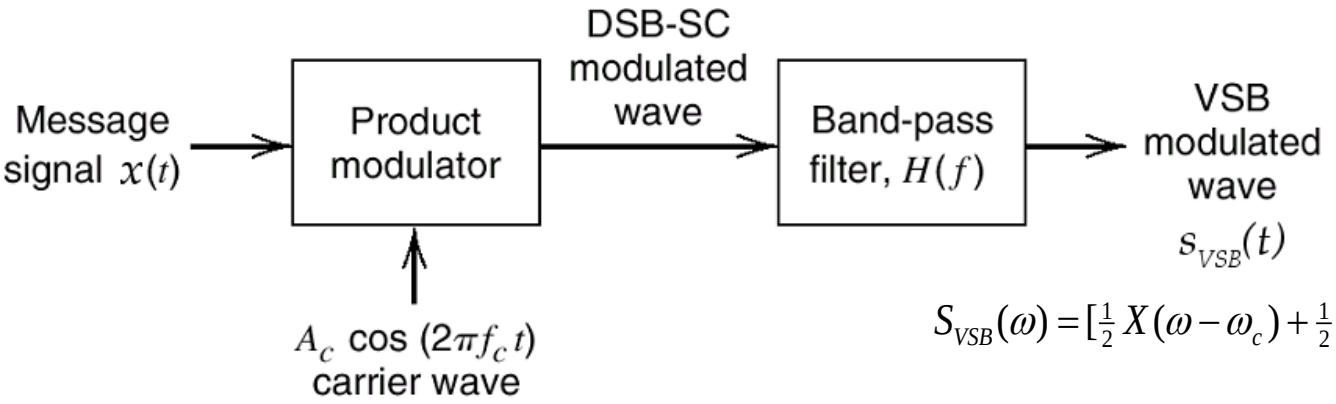
The partial suppression of one sideband reduces the required bandwidth from that required for DSB but does not match the spectrum efficiency of SSB.

If a large carrier is also transmitted, the desired signal can be recovered using an envelope detector.

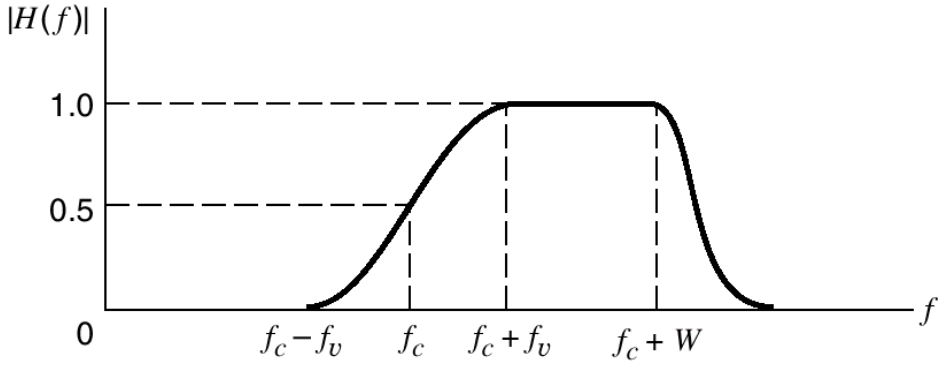
If no carrier is sent, the signal can be recovered using a synchronous detector or the injected carrier method.

Amplitude Modulation: VSB

Generation of VSB Signals



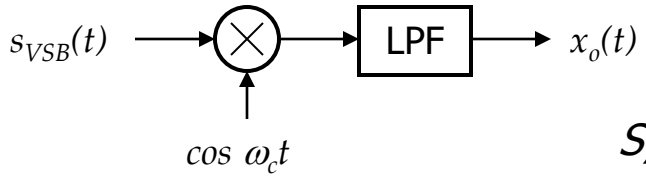
$$S_{VSB}(\omega) = [\tfrac{1}{2} X(\omega - \omega_c) + \tfrac{1}{2} X(\omega + \omega_c)]H(\omega)$$



The filtering operation can be represented by a filter $H(f)$ that passes some of the lower (or upper) sideband and most of the upper (or lower) sideband.

Amplitude Modulation: VSB

Demodulation of VSB Signals



Synchronous Demodulation

The spectral density of the received vestigial-sideband signal is

$$S_{VSB}(\omega) = [\tfrac{1}{2} X(\omega - \omega_c) + \tfrac{1}{2} X(\omega + \omega_c)]H(\omega)$$

The output of the synchronous detector is

$$x_o(t) = [s_{VSB}(t)\cos\omega_c t]_{LP}$$

$$X_o(\omega) = \tfrac{1}{4} X(\omega)\left[H(\omega + \omega_c)\right]_{LP} + \tfrac{1}{4} X(\omega)\left[H(\omega - \omega_c)\right]_{LP}$$

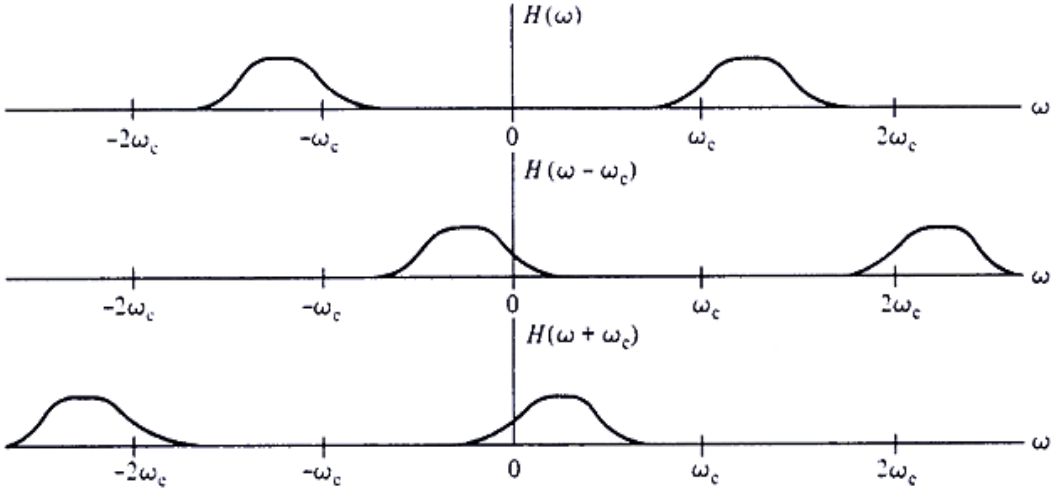
$$= \tfrac{1}{4} X(\omega)\left[H(\omega + \omega_c) + H(\omega - \omega_c)\right]_{LP}$$

For faithful reproduction of $x(t)$, we require that

$$[H(\omega - \omega_c) + H(\omega + \omega_c)]_{LP} = constant, \quad |\omega| \le \omega_m .$$

Amplitude Modulation: VSB

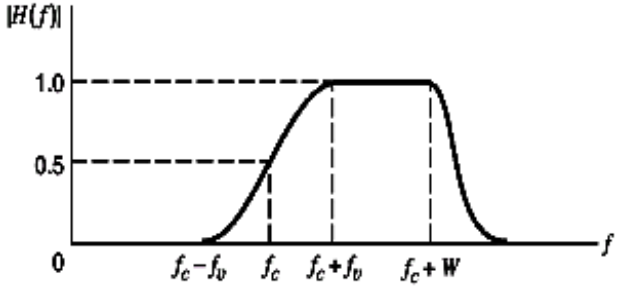
$$[H(\omega-\omega_c)+H(\omega+\omega_c)]_{LP}=constant, \quad |\omega|\le\omega_m.$$



By letting the constant be $2H(\omega_c)$:

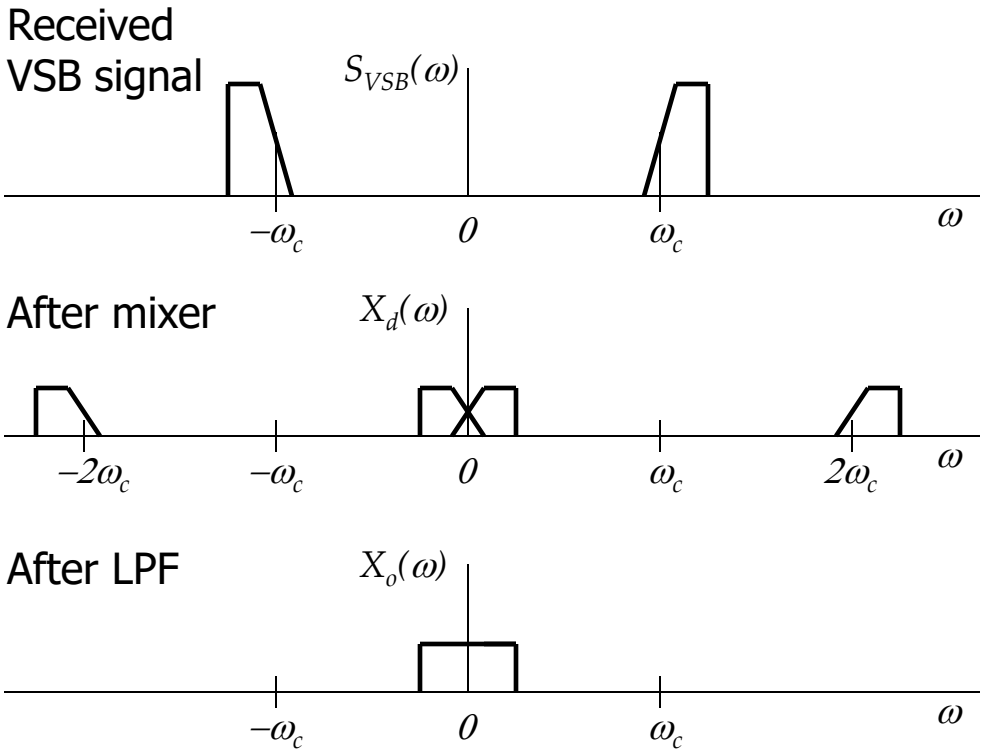
$$[H(\omega-\omega_c)+H(\omega+\omega_c)]_{LP}=2H(\omega_c), \quad |\omega|\le\omega_m$$

Thus, $H(\omega)$ exhibits **odd symmetry** around the carrier frequency ω_c.
The sum of the values of $H(\omega)$ at any two frequencies equally displaced above and below ω_c is unity.



Amplitude Modulation: VSB

Synchronous Demodulation of VSB Signals

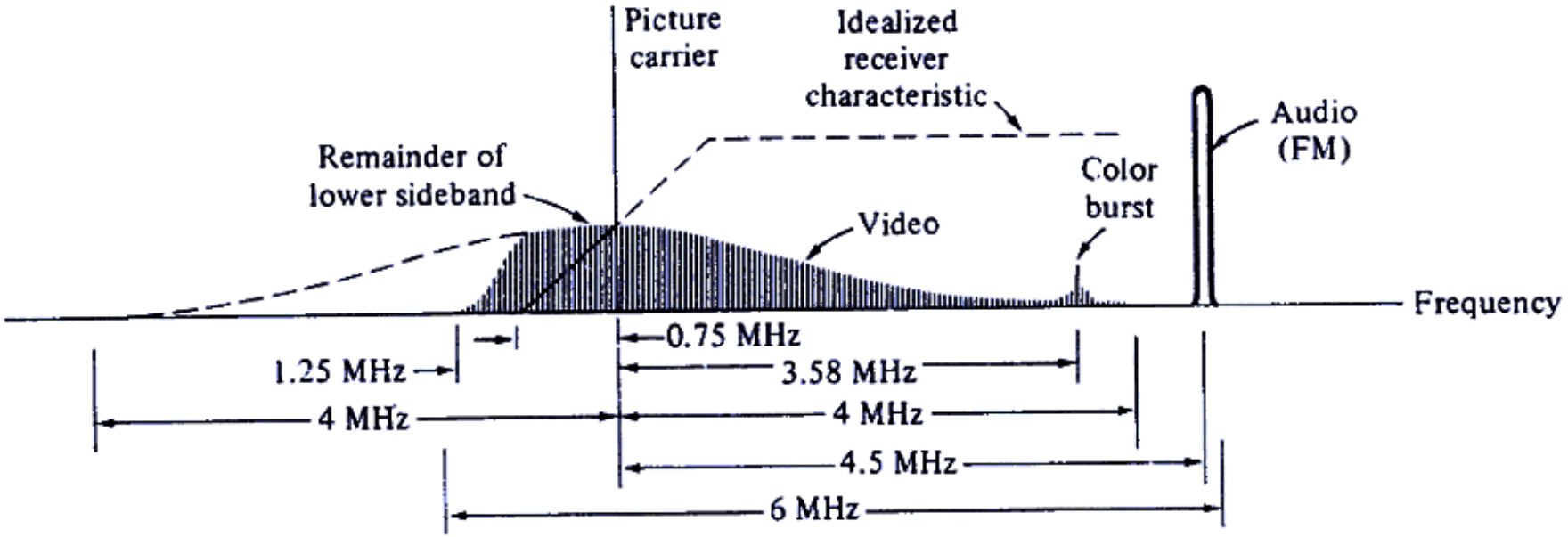


Amplitude Modulation: VSB

Example of VSB Signals: ***Television signal***

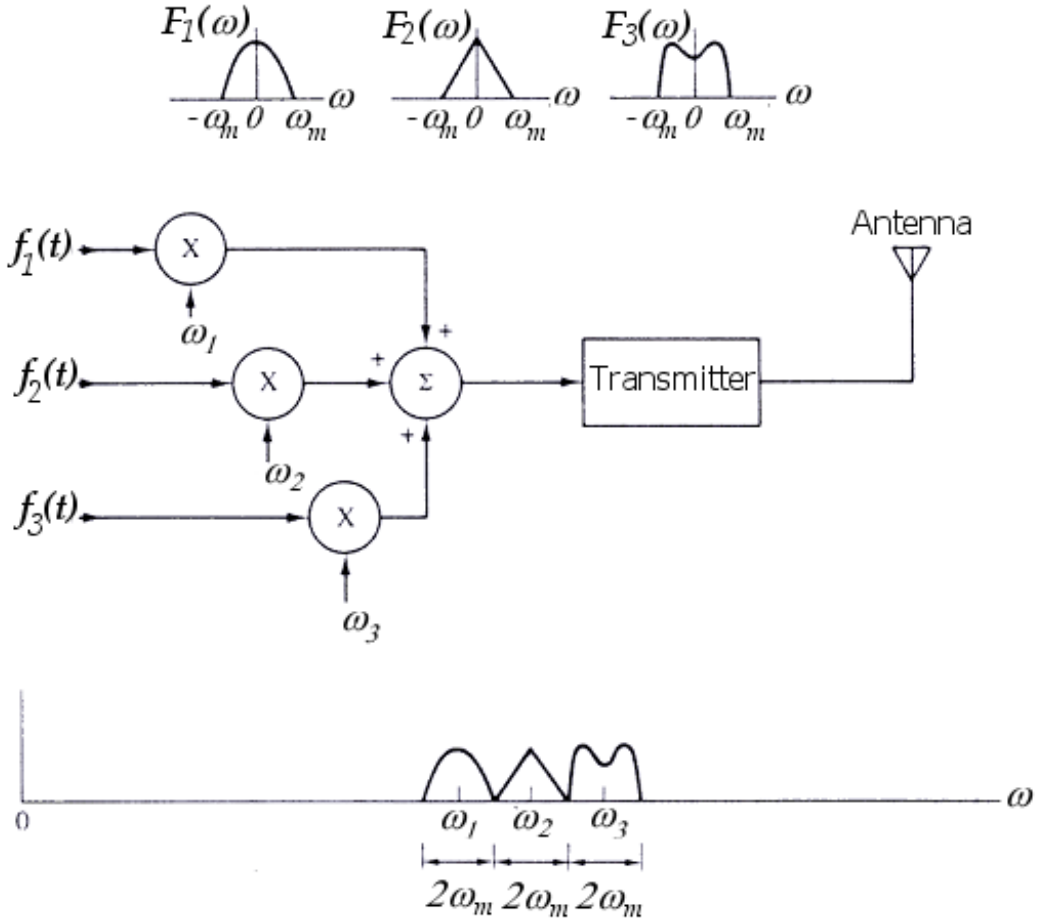
Television picture signal has nominal bandwidth of 4.5MHz

If DSB modulation is used, it requires at least 9MHz for each TV channel. So, VSB modulation is used so that the whole TV signal is confined to about 6MHz.

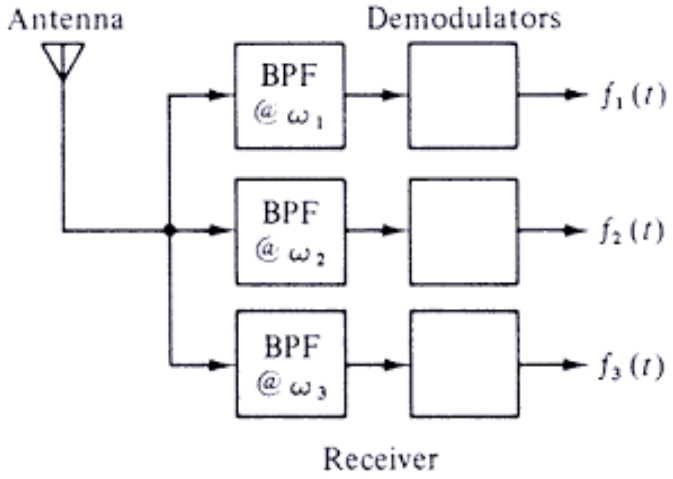


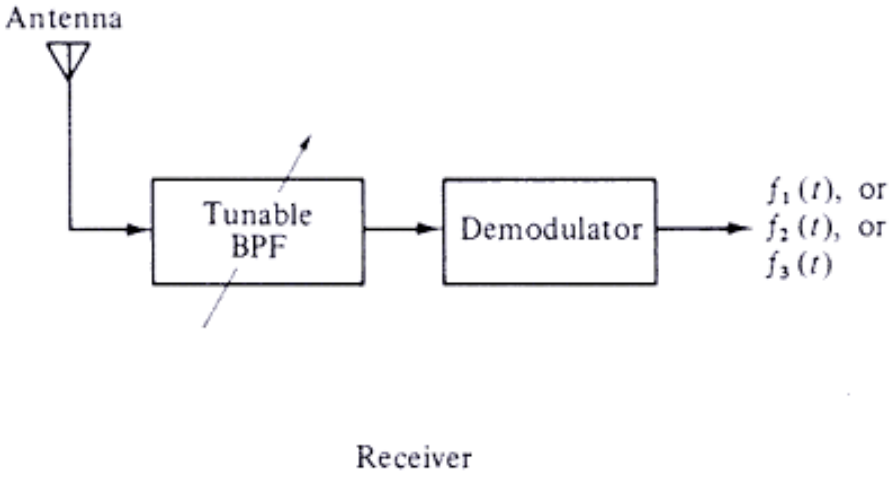
Frequency Division Multiplexing (FDM)

Frequency-division multiplexing is the positioning of signal spectra in frequency such that each signal spectrum can be separated out from all the others by filtering.



Frequency Division Multiplexing (FDM)





Example: commercial radio and television receiver

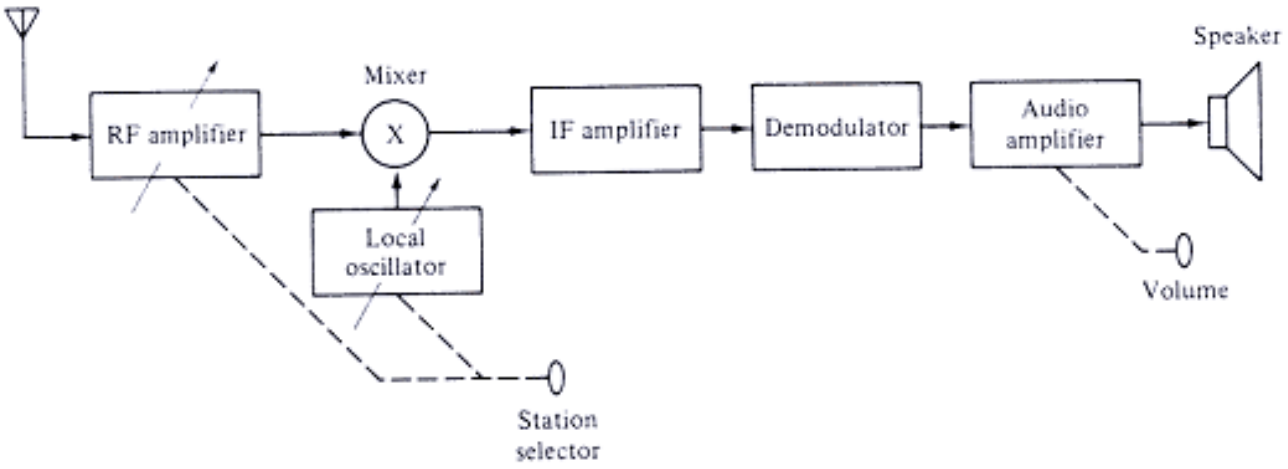
Intermediate Frequency (IF)

Heterodyning means the translating or shifting in frequency.

In the heterodyne receiver the incoming modulated signal is translated in frequency, thus occupying an equal bandwidth centered about a new frequency, known as an intermediate frequency (IF), which is fixed and is not dependent on the received signal center frequency.

The signal is amplified at the IF before demodulation.

If this intermediate frequency is lower than the received carrier frequency but above the final output signal frequency, it is called a superheterodyne receiver.



Intermediate Frequency (IF)

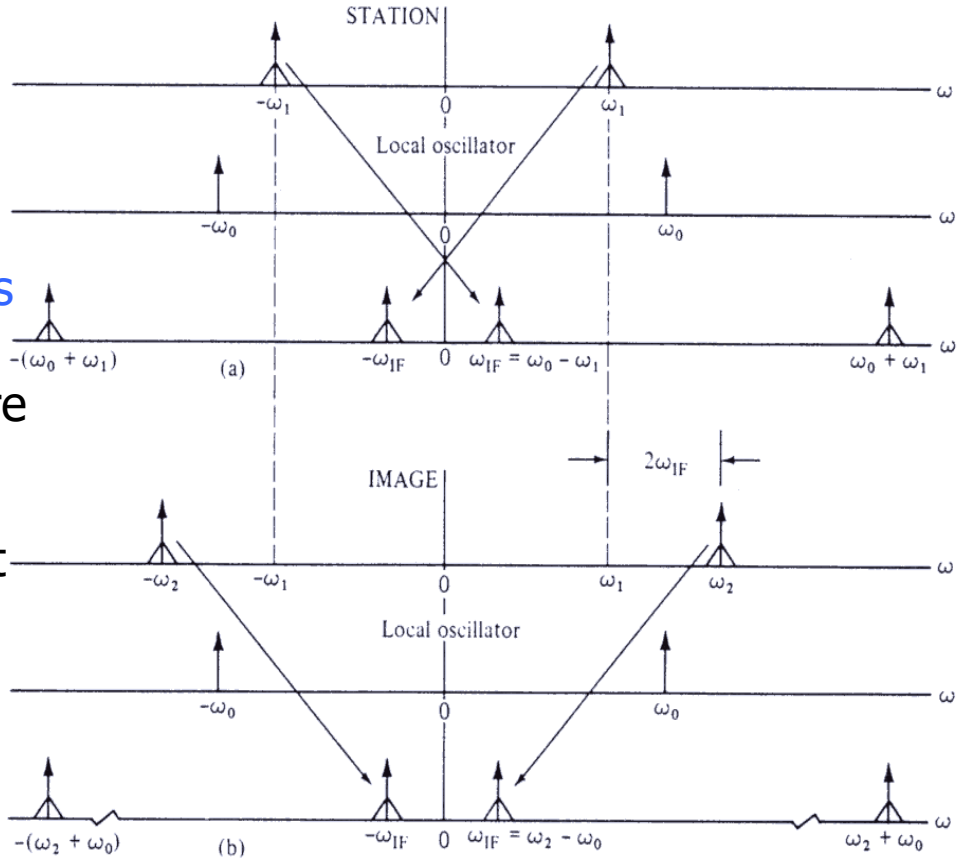
Advantage:
The amplification and filtering is performed at a fixed frequency regardless of station selection.

Disadvantage:
Image-frequency problem

Two ways to solve this problem

i. Choose the intermediate frequency as high as possible and practical.
ii. Attenuate the image frequency before heterodyning.

The intermediate frequency chosen must be free from other strong transmissions or otherwise the receivers will amplify these spurious signals as they leak into the high-gain IF stages.



Intermediate Frequency (IF)

Example:

A given radar receiver operating at a frequency of 2.80 GHz and using the super-heterodyne principle has a local oscillator frequency of 2.86 GHz . A second radar receiver operates at the image frequency of the first and interference results.

(a) Determine the intermediate frequency of the first radar receiver.

(b) What is the carrier frequency of the second receiver?

(c) If you were to redesign the radar receiver, what is the minimum intermediate frequency you would choose to prevent image-frequency problems in the 2.80-3.00 GHz radar band?

Solution:

(a) $f_{IF} = f_{LO} - f_c = 2.86\text{GHz} - 2.80\text{GHz} = 60\text{MHz}.$

(b) $f_{IMAGE} = f_c + 2f_{IF} = 2.80\text{GHz} + 0.12\text{GHz} = 2.92\text{GHz}.$

(c) $2f_{IF} \geq f_{MAX} - f_{MIN} = 3.00\text{GHz} - 2.80\text{GHz} = 0.20\text{GHz};\ f_{IF} \geq 100\text{MHz}.$

Intermediate Frequency (IF)

